Predictive Stochastic Modeling for Financial Markets

Revised edition

Jaden Fix

Original thesis advisor: Jonathan Kaplan

This revised edition of the undergraduate thesis that received the Adam Smith Award rebuilds the analysis from Robert Shiller's primary data, evaluates bear-market forecasts in real time, adds time-varying-transition regime models, machine-learning classifiers, dynamic forecast combination and multiple-comparison inference, and reports a broad set of robustness experiments. A single script produces every estimate, table and figure. The award version (February 2025), its R code and the code for this edition are available at github.com/jadenfix/Predictive-Stochastic-Modeling-for-Financial-Markets-.

Abstract

Can market and macroeconomic indicators forecast bear markets in the S&P 500? Using Robert Shiller's monthly data from 1871 to June 2026, I date 25 bear markets in the real index with a 20% threshold rule and evaluate probability forecasts made every month from January 1960 to May 2026, with every model re-estimated on the information available at the forecast origin. Returns alternate between a calm bull regime (mean 0.88% a month, expected duration 16.6 months) and a volatile bear regime (mean −1.75%, 5.0 months). Letting the regime transition probabilities depend on inflation, valuation, volatility and recent drawdowns improves the in-sample fit by the likelihood ratio (51.0, $p < 0.001$), though not by BIC, and does not improve out-of-sample accuracy. At one month, a logit on six state variables reduces the Brier score by 25.4% relative to the unconditional bear frequency, against 17.6% for simply knowing whether a confirmed bear market is under way; gradient boosting (26.8%) and dynamic model averaging (23.4%) do about as well. These gains over persistence do not survive a realistic account of uncertainty: loss differences are strongly autocorrelated, Hansen's test for superior predictive ability across the 24 competing models gives $p = 0.189$ at one month and 0.468 across all horizons, and persistence remains in the model confidence set at every horizon. Beyond six months no model reliably beats the base rate. Bear markets are partly recognizable in real time; with these data they are not predictable months in advance.

1 Introduction

Bear markets destroy a large share of household and institutional wealth in a short time, and whether they can be anticipated is a direct test of how much public information stock prices already contain. Two literatures bear on the question. One finds that equity returns alternate between persistent bull and bear regimes (Hamilton, 1989; Maheu and McCurdy, 2000; Ang and Timmermann, 2012). The other finds that valuation ratios and macroeconomic conditions carry information about future returns and downturns (Campbell and Shiller, 1988; Fama and French, 1988; Chen, 2009). If regimes are persistent and their onset is related to observable conditions, bear markets should be at least partly forecastable.

This paper asks how much of that forecastability survives a strict real-time test. A forecast may use only information that was published when it was made, every model is re-estimated at every forecast origin, and accuracy is measured out of sample against two baselines: the unconditional frequency of bear markets and the persistence of the current, observable market state. The second baseline is the demanding one. A bear market dated from its peak is only recognized once prices have fallen far enough, so a model that merely notices an ongoing decline will look skillful against the base rate without telling an investor anything new.

The analysis proceeds in five steps. First, I date bear markets in Shiller's real S&P 500 series from 1871 to 2026 with an explicit threshold rule in the spirit of Lunde and Timmermann (2004), so that the forecast target is an observable event rather than a model output. Second, I estimate Markov-switching models of real returns, including a model whose transition probabilities depend on lagged predictors (Diebold, Lee and Weinbach, 1994; Filardo, 1994). Third, I estimate logit and probit models (Estrella and Mishkin, 1998; Chen, 2009), an elastic-net logit and gradient-boosted trees in the spirit of Gu, Kelly and Xiu (2020), and equal-weight and dynamic combinations of the individual forecasts (Rapach, Strauss and Zhou, 2010; Raftery, Kárný and Ettler, 2010). Fourth, I evaluate the real-time forecasts over 1960:02-2026:06 at horizons of 1, 3, 6, 12 months with proper scoring rules, the Murphy (1973) decomposition, and inference that accounts for the number of models compared: the test for superior predictive ability of Hansen (2005), the model confidence

set of Hansen, Lunde and Nason (2011) and the conditional test of Giacomini and White (2006). Fifth, I vary the dating rule, the regime specification, the estimation window, the evaluation period and the predictor set.

The evidence supports four findings. Regime structure in returns is strong: a calm bull regime and a short, volatile bear regime appear in every specification and subsample, and time-varying transition probabilities fit the data better in sample. Inflation, the dividend yield and weak recent returns are associated with bear markets in sample, and real-time models beat the unconditional frequency at horizons of one to three months. Much of that short-horizon skill comes from recognizing a bear market that has already begun; the best models improve on persistence by several points of Brier skill at one month, mainly through better calibration, but the improvement is within sampling error once the dependence of the forecast errors and the number of models are taken into account. Beyond six months nothing reliably improves on the base rate, and the flexible models do worst there.

The paper proceeds as follows. Section 2 reviews related work, Section 3 describes the data and the dating rule, Section 4 the models, the real-time design and the inference, Section 5 the main results and Section 6 the robustness experiments. Section 7 discusses limitations and Section 8 concludes.

2 Related literature

Regime-switching models in the tradition of Hamilton (1989) describe the stock market as moving between a high-return, low-volatility state and a low-return, high-volatility state. Maheu and McCurdy (2000) identify bull and bear markets this way, and Ang and Timmermann (2012) survey the evidence that regime changes are a pervasive feature of financial markets. Diebold, Lee and Weinbach (1994) and Filardo (1994) let the transition probabilities depend on observable variables, which turns a descriptive regime model into a potential forecasting model. Rule-based dating is the main alternative to model-based regimes: Pagan and Sossounov (2003) adapt business-cycle turning-point algorithms to equity prices, and Lunde and Timmermann (2004) define bull and bear markets by threshold movements from the latest peak or trough. Chen (2009) and Chen, Chen and Chou (2017) use such dates as the dependent variable in probit models and find that macroeconomic and credit variables help predict bear markets.

On the predictor side, Campbell and Shiller (1988) and Fama and French (1988) document that dividend yields forecast long-horizon returns, while Goetzmann and Jorion (1993), Stambaugh (1999) and Lewellen (1999) show that inference with persistent, price-scaled predictors is delicate. Pontiff and Schall (1998) report related evidence for book-to-market ratios, and Hammerschmid and Lohre (2018) find that return predictability itself varies across regimes. Welch and Goyal (2008) show that many predictors that look strong in sample fail out of sample, Rapach, Strauss and Zhou (2010) find that combining forecasts delivers more reliable gains than any single predictor, and Gu, Kelly and Xiu (2020) show that regularized and tree-based machine-learning methods can extract more predictive information from many signals than linear regressions. A study that compares many models also needs inference that accounts for the search: White's (2000) Reality Check, Hansen's (2005) test for superior predictive ability and the model confidence set of Hansen, Lunde and Nason (2011) are designed for that problem. This paper applies these out-of-sample standards to the prediction of bear markets rather than of average returns.

3 Data

3.1 Source, vintage and timing

The data are Robert Shiller's monthly U.S. stock market series (Shiller, 2015), the workbook "U.S. Stock Markets 1871-Present and CAPE Ratio" linked from his Yale data page and distributed at shillerdata.com. I downloaded the file `ie_data.xls` on 18 October 2026 (the server reported it as last modified on 6 October 2026; 1.7 MB; the SHA-256 hash is given in Appendix B). Every number in this paper is computed from the "Data" sheet of this file: the S&P Composite price (a monthly average of daily closes), dividends and earnings per share, the consumer price index, the long-term interest rate, Shiller's real total-return-scaled earnings and his real total return index of ten-year Treasury bonds. The workbook runs from January 1871 to October 2026, but dividends and earnings end earlier and the most recent price, CPI and yield observations are provisional (the sheet notes: "Oct price is Oct 2nd close; Sept/Oct CPI estimated; Oct GS10 is Oct 2nd value"). The sample therefore ends in June 2026, the last month in which every series is reported, and contains 1,865 monthly returns from February 1871.

The award version of this thesis used `one45.csv`, an extract of an earlier vintage of the same workbook covering 1900 to early 2024. On the 1,491 overlapping months the two files agree to the archived file's two decimals except for 17 observations, all in the archived file's final months (October 2023 onward), where it held estimates that Shiller has since revised by at most 1.0% (Appendix B). The substantive change is that the current vintage adds the years 1871 to 1899 and 2024 to 2026.

I treat the end of month t as the forecast origin and impose publication lags on everything that is not known then: the consumer price index enters with a lag of 1 month, and dividends and earnings, which Shiller interpolates from quarterly reports (annual before 1926), enter with a lag of 6 months. Real values are obtained by deflating with the CPI available at t. Shiller's monthly bond return for month t uses the following month's yield, so it is never used directly; his real bond index at t compounds only earlier returns, which the code verifies. Table 1 lists the predictors, their construction and their publication lags.

The award version's regime model and forecast targets relied on columns of `one45.csv` that held future rather than past prices (the column labeled `realpricelag` equals the following month's real price on 1,490 of 1,490 comparable rows) and on "ahead" targets that were shifted backward in time. This edition builds every predictor from data dated at or before the forecast origin and verifies with automated tests that no predictor or forecast changes when data after its origin are altered; Appendix A relates the award version's estimates to the ones reported here.

3.2 Dating bear markets

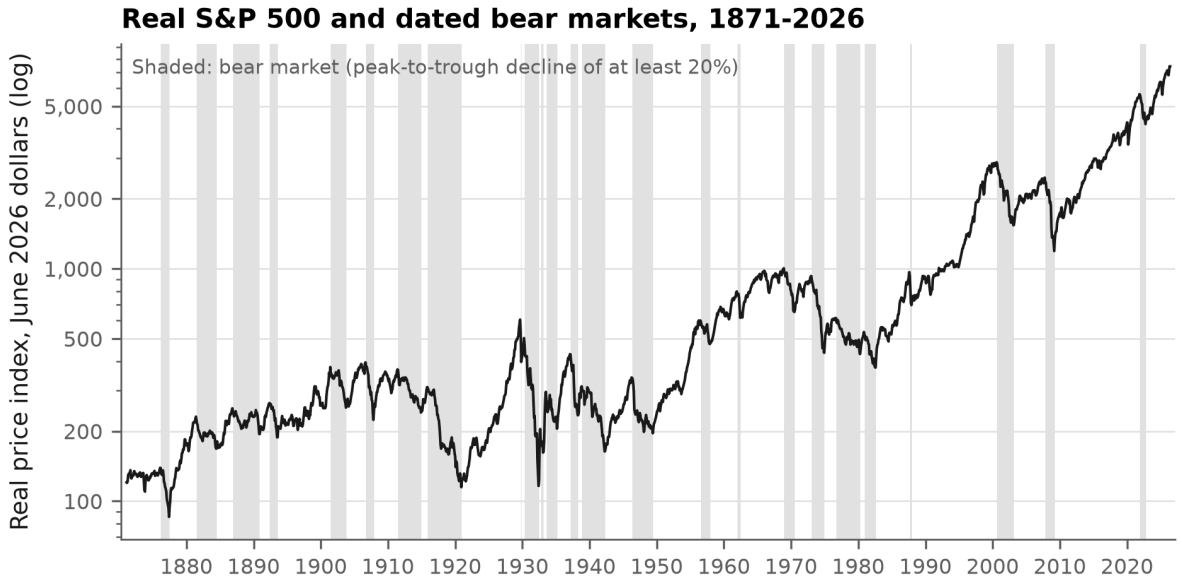
I date bear markets on the real price index with a symmetric threshold rule in the spirit of Lunde and Timmermann (2004). Starting from a bull state, the running maximum is a candidate peak; when the index falls 20% below it, the peak is confirmed and a bear market begins in the following month. In the bear state the running minimum is a candidate trough; when the index rises 20% above it, the trough is confirmed and the bear market ends at the trough month. The rule yields 25 bear markets since 1871, 9 of them beginning after 1960 (Table 2, Figure 1); the most recent ran from November 2021 to October 2022. The median bear market lasts 20 months with a median peak-to-trough decline of 34.2%. Bear months account for 32.0% of the full sample and 21.8% of the months since 1960.

Table 1: Predictors, construction and publication lags

Predictor	Construction	Units	Lag (months)	Mean	SD
Monthly-change predictors (award-version variables; appendix)					
CPI inflation, monthly	100 dlog CPI	% per month	1	0.30	0.36
Change in 10-year yield	Monthly change	pp	0	0.00	0.27
Real bond return	100 dlog real bond index	% per month	0 (CPI 1)	0.17	1.97
Real dividend growth	100 dlog real dividends	% per month	6	0.17	0.62
Real total return	Price return plus D/12P, less inflation	% per month	0 (D 6, CPI 1)	0.53	3.63
TR-scaled earnings growth	100 dlog real TR-scaled earnings	% per month	6	0.46	4.75
Change in log TR-CAPE	100 dlog TR price / 10-yr mean earnings	% per month	0 (E 6)	0.07	3.64
State predictors (main bear-market models)					
Log dividend yield	log(D / P)	log points	0 (D 6)	−3.69	0.42
Log CAPE	log(P / 10-yr mean real earnings)	log points	0 (E 6)	3.00	0.44
CPI inflation, 12-month	100 x 12-month dlog CPI	%	1	3.64	2.63
12-month change in 10-year yield	12-month change	pp	0	0.00	1.11
12-month real price change	100 x 12-month dlog real price	%	0 (CPI 1)	3.55	15.87
12-month realized volatility	sd of monthly real returns x sqrt(12)	% annualized	0 (CPI 1)	11.23	4.86
Additional state predictors (predictor experiments, rich machine-learning set)					
Excess CAPE yield	100 E10/P less (yield less 10-yr inflation)	%	0 (E 6, CPI 1)	3.25	2.35
Log payout ratio	log(D / E)	log points	6	−0.77	0.30
12-month real earnings growth	100 x 12-month dlog real earnings	%	6	2.86	34.66
10-year Treasury yield	Level	%	0	5.76	2.91
Drawdown from 12-month high	100 x log(P / 12-month high)	%	0 (CPI 1)	−6.48	9.74

Notes: Statistics over January 1960 to June 2026. A forecast is made at the end of month t. Lag gives the publication lag applied to the inputs: CPI enters with one month, dividends (D) and earnings (E) with six. Real values are deflated by the CPI available at t.

The dated labels are ex post: a peak is only known to have been a peak once the 20% decline has occurred, which took a median of 11 months. The same rule run on data through month t



Source: R. J. Shiller, U.S. Stock Markets 1871-Present and CAPE Ratio (shillerdata.com, downloaded 18 October 2026); author's calculations.

Figure 1: Real S&P 500 price index (log scale) with dated bear markets shaded.

Table 2: Dated bear markets in the real S&P 500, 1871-2026

Peak	Trough	Months	Decline (%)	Confirmed	Months to confirm
1876-02	1877-06	16	−38.5	1877-01	11
1881-06	1884-06	36	−27.1	1882-05	11
1886-11	1890-11	48	−22.5	1890-11	48
1892-05	1893-07	14	−28.8	1893-05	12
1901-06	1903-10	28	−33.1	1903-06	24
1906-09	1907-11	14	−43.4	1907-03	6
1911-06	1914-12	42	−34.8	1913-05	23
1915-11	1920-12	61	−62.9	1917-05	18
1929-09	1929-11	2	−34.2	1929-11	2
1930-04	1932-06	26	−76.9	1930-10	6
1932-09	1933-02	5	−20.8	1933-02	5
1933-07	1935-03	20	−30.6	1934-07	12
1937-02	1938-04	14	−45.4	1937-09	7
1938-11	1942-04	41	−47.5	1940-06	19
1946-04	1949-06	38	−42.4	1946-09	5
1956-07	1957-12	17	−20.8	1957-11	16
1961-12	1962-06	6	−23.0	1962-06	6
1968-12	1970-07	19	−35.1	1970-01	13
1973-01	1974-12	23	−53.3	1973-12	11
1976-09	1980-04	43	−30.0	1978-01	16
1980-11	1982-08	21	−29.7	1982-01	14
1987-08	1987-12	4	−27.9	1987-11	3
2000-08	2003-02	30	−46.4	2001-03	7
2007-10	2009-03	17	−51.7	2008-07	9
2021-11	2022-10	11	−25.6	2022-06	7

Notes: A bear market runs from a peak to the following trough when the real index falls at least 20% below the peak before rising 20% above the trough. 'Confirmed' is the first month in which the 20% decline had occurred, that is, when the bear market became knowable in real time.

gives the real-time state, which is observable at t, and "vintage" labels for all earlier months. The out-of-sample exercise trains every model on vintage labels only.

4 Methodology

4.1 Markov-switching models of returns

Let $r_t = 100 \cdot \Delta \ln P_t$ be the monthly real return. The main regime model is a two-state Markov-switching autoregression with switching mean and variance (Hamilton, 1989):

$$r_t - \mu_{S_t} = \varphi\big(r_{t-1} - \mu_{S_{t-1}}\big) + \sigma_{S_t}\varepsilon_t, \quad \varepsilon_t \sim N(0,1), \quad S_t \in \{\text{bull}, \text{bear}\}, \tag{1}$$

where S_t follows a first-order Markov chain with transition probabilities $p_{ij} = \Pr(S_t = j \mid S_{t-1} = i)$. Setting $\varphi = 0$ gives the switching mean-variance model. The bear regime is the one with the lower mean, and the expected duration of regime i is $1/(1-p_{ii})$. The Hamilton filter delivers the filtered probability $Q_t = \Pr(S_t = \text{bear} \mid r_1, ..., r_t)$, which uses only returns through t and is a legitimate real-time quantity once the parameters are fixed. The smoothed probability conditions on the whole sample and is used only for description. An h-step regime forecast is $\Pr(S_{t+h} = \text{bear} \mid r_1, ..., r_t) = [1 - Q_t, Q_t]\boldsymbol{P}^h e_2$, where $\boldsymbol{P}$ is the transition matrix.

Time-varying transition probabilities. Following Diebold, Lee and Weinbach (1994) and Filardo (1994), the TVTP model replaces the constant transition probabilities by

$$\Pr(S_t = \text{bull} \mid S_{t-1} = i, z_{t-1}) = \Lambda(z_{t-1}^\top b_i), \quad i \in \{\text{bull}, \text{bear}\}, \tag{2}$$

where Λ is the logistic function and z_{t-1} contains a constant and standardized predictors observed at $t-1$: the log dividend yield, twelve-month inflation, twelve-month realized volatility and the drawdown from the twelve-month high. The model nests the fixed-transition MS-AR(1), so the

likelihood-ratio test of zero slopes is chi-squared with two degrees of freedom per predictor. An h-step forecast from t applies the transition matrix implied by z_t h times, because later values of the predictors are unknown at t.

4.2 Binary models for dated bear markets

The dated bear indicator y_{t+h} is the forecast target. The binary regression

$$\Pr(y_{t+h} = 1 \mid x_t) = F(\alpha + x_t^\top \beta) \tag{3}$$

uses the logistic (logit) or standard normal (probit) distribution function F. The 6 main predictors in x_t are the log dividend yield, the log CAPE ratio with lagged earnings, twelve-month CPI inflation, the twelve-month change in the ten-year yield, the twelve-month real price change and twelve-month realized volatility. In-sample estimates use Newey and West (1987) standard errors, since overlapping horizons make the errors serially correlated.

4.3 Machine learning

Two classifiers in the spirit of Gu, Kelly and Xiu (2020) estimate the same conditional probability with more flexible functional forms. The elastic-net logit (Zou and Hastie, 2005) penalizes a mixture of the absolute and squared coefficients, with inverse penalty $C \in \{0.01, 0.1, 1, 10\}$ and mixing weight in $\{0.1, 0.5, 0.9\}$ (12 combinations). Gradient-boosted classification trees (Friedman, 2001) use depths $\{2, 3\}$ and $\{100, 300\}$ boosting rounds with a learning rate of 0.05 (4 combinations). Hyperparameters are chosen without looking outside the training window: the last 10 years of training origins form a validation block, the estimation block ends h months before it so that no estimation target falls in the validation period, and the combination with the lowest validation Brier score is refit on the whole window. Each method is applied to the six main predictors and to a rich set that adds the 5 additional predictors of Section 6 and the real-time bear state.

4.4 Forecast combination

The equal-weight combination averages the probability forecasts of univariate logits on each of the 11 predictors (Rapach, Strauss and Zhou, 2010). Dynamic model averaging (Raftery, Kárný and Ettler, 2010) instead weights forecasts by their recent accuracy. With a fixed set of member forecasts, the DMA recursion, which flattens the previous weights by a forgetting factor α and then multiplies them by each member's predictive likelihood, gives weights

$$w_{k,t} \propto \exp\left(\sum_s \alpha^{t-h-s} \ln p_k(y_{s+h})\right), \tag{4}$$

where the sum runs over earlier forecasts whose outcomes y_{s+h} are known at t, scored against the vintage labels available at t. I use $\alpha = 0.99$ and report $\alpha \in \{0.95, 0.99, 1\}$ as a check; $\alpha = 1$ is recursive Bayesian model averaging. DMA is applied to the univariate logits and to the main model set.

4.5 Real-time design

Forecasts are made at every month-end from January 1960 to May 2026 for horizons of 1, 3, 6, 12 months. At each origin t every model is re-estimated on an expanding window using only (i) predictors dated $s \le t$; (ii) vintage bear labels from dating the price history through t, so that a training pair (x_s, y_{s+h}) requires $s + h \le t$; and (iii) for the regime models, parameters re-estimated each January on all returns from 1871 through the origin (67 estimations of each model) with filtered probabilities from those parameters. The logits and probits are re-estimated every month.

The elastic net and the boosted trees are re-estimated, with hyperparameter selection, each January and applied to the current predictors every month, which keeps their cost modest. Training starts in June 1881, when all predictors are first available. The two baselines are the **unconditional frequency**, the training-sample share of bear months h months ahead, and **persistence**, the training-sample frequency of a bear market h months ahead conditional on the current real-time state (in a confirmed bear market or not). The regime-model forecasts are logit recalibrations of the h-step regime forecast, which map the regime probability into the probability of a dated bear market on training data only.

4.6 Evaluation and inference

Accuracy is measured by the Brier score (Brier, 1950), the log loss (with probabilities truncated to $[0.001, 1 - 0.001]$), the area under the ROC curve (AUC) and calibration curves. Brier skill is the percentage reduction in the Brier score relative to the unconditional frequency. The Murphy (1973) decomposition splits the Brier score into reliability (miscalibration, lower is better), resolution (how much the outcome frequency varies with the forecast, higher is better) and the uncertainty of the outcome, using 10 equal-width forecast bins.

Pairwise differences in Brier score are tested with the Diebold and Mariano (1995) statistic with the small-sample correction of Harvey, Leybourne and Newbold (1997). The conventional Bartlett window of $h - 1$ lags (implemented as $\max(h - 1, 1)$, so one lag at one month) is too short here: bear markets last many months, so a model's advantage over persistence persists too: at one month the first-order autocorrelation of the loss differences against persistence has a median of 0.85 across models. I therefore report the tests with the conventional window and with 23 lags. Because many models and horizons are compared, three further procedures control for the search. Hansen's (2005) test for superior predictive ability (SPA) asks whether any model beats persistence, with White's (2000) Reality Check as its conservative bound; the joint version treats every model-horizon pair as a separate forecast. The model confidence set (MCS) of Hansen, Lunde and Nason (2011) is the set of models that cannot be distinguished from the best at the 10% level. The conditional predictive ability test of Giacomini and White (2006) asks whether the best model's advantage over persistence depends on the real-time bear state. All bootstrap procedures resample target months in circular blocks of 24 months (Künsch, 1989) with 2,000 draws, jointly for all models.

5 Results

5.1 Returns have two persistent regimes

Table 3 reports the fixed-transition regime models. In the MS-AR(1) model for January 1960 to June 2026, the bull regime has a mean real return of 0.88% per month with a standard deviation of 2.41%, while the bear regime has a mean of −1.75% and a standard deviation of 5.57%. Both regimes are persistent ($p_{\text{bull,bull}} = 0.940$, $p_{\text{bear,bear}} = 0.798$), implying expected durations of 16.6 months for bull regimes and 5.0 months for bear regimes. The autoregressive term ($\varphi = 0.196$) partly reflects the time averaging of monthly prices and lowers the AIC from 4,174.3 to 4,141.0, so the MS-AR(1) is the main specification. On the full 1871 to 2026 sample the bull regime is more persistent (expected duration 34.2 months) and the bear regime more volatile (standard deviation 7.34%).

Figure 2 contrasts the filtered and smoothed bear-regime probabilities. The smoothed series marks the dated bear markets more cleanly (AUC 0.860 against 0.802 for the contemporaneous dated state), but that advantage comes from conditioning on later returns. The filtered series is

Table 3: Two-regime Markov-switching models of monthly real S&P 500 returns

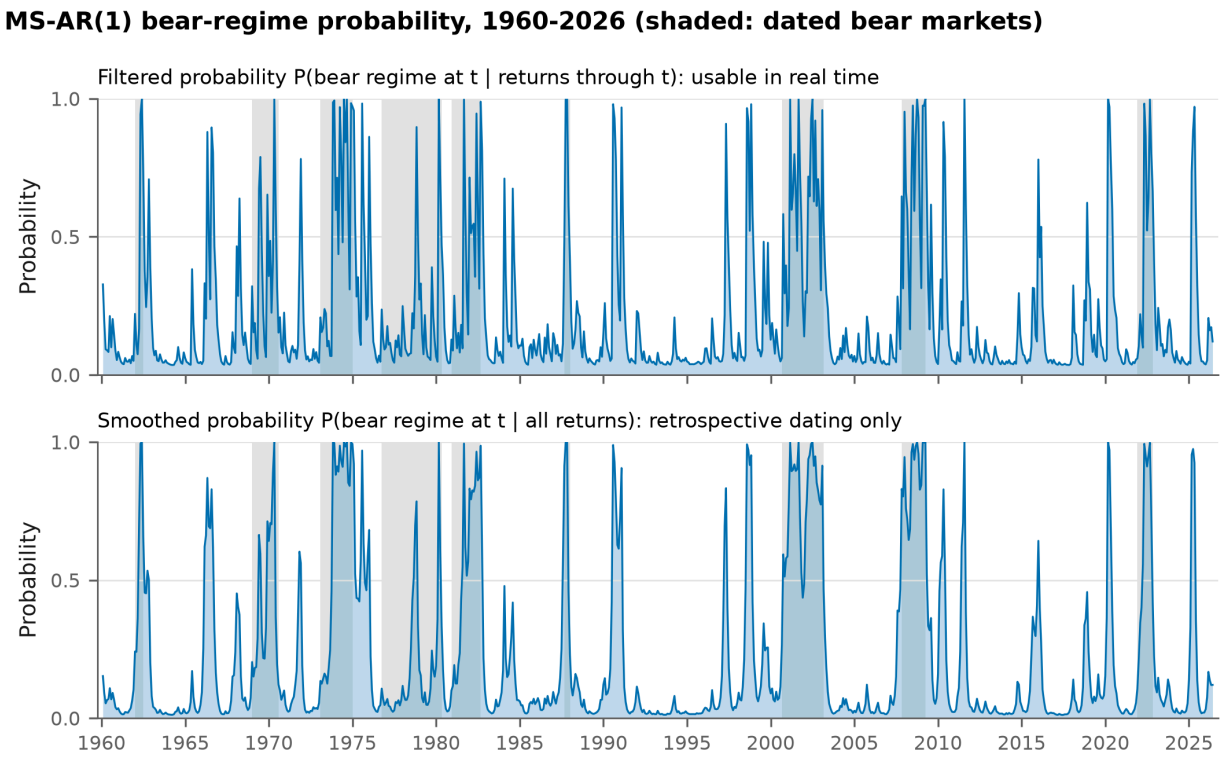
	MS, 1960-2026	MS-AR(1), 1960-2026	MS, 1871-2026	MS-AR(1), 1871-2026
Mean monthly real return (%)				
Bull regime	0.953***	0.876***	0.703***	0.626***
	(0.128)	(0.141)	(0.086)	(0.104)
Bear regime	−1.841***	−1.749***	−2.052***	−2.024***
	(0.562)	(0.637)	(0.480)	(0.601)
Innovation variance				
Bull regime	6.18	5.82	8.32	8.04
	(0.54)	(0.53)	(0.43)	(0.43)
Bear regime	30.31	31.05	53.51	53.81
	(4.09)	(4.32)	(5.50)	(5.94)
Bull volatility (sd, %)	2.49	2.41	2.88	2.84
Bear volatility (sd, %)	5.51	5.57	7.31	7.34
Autoregression				
AR(1) coefficient		0.196***		0.233***
		(0.035)		(0.023)
Transitions				
P(bull to bull)	0.945	0.940	0.970	0.971
	(0.017)	(0.020)	(0.007)	(0.008)
P(bear to bear)	0.818	0.798	0.859	0.851
	(0.055)	(0.065)	(0.031)	(0.036)
Expected bull duration (months)	18.1	16.6	33.6	34.2
Expected bear duration (months)	5.5	5.0	7.1	6.7
Fit				
Log-likelihood	−2,081.2	−2,063.5	−5,060.7	−5,010.4
AIC	4,174.3	4,141.0	10,133.4	10,034.7
BIC	4,202.4	4,173.7	10,166.6	10,073.5
Observations	798	797	1,864	1,863
Sample	1960:01-2026:06	1960:02-2026:06	1871:03-2026:06	1871:04-2026:06

Notes: Maximum-likelihood estimates; standard errors in parentheses from the numerical Hessian. The bear regime is the regime with the lower mean. Switching mean and variance; the AR(1) coefficient is common to both regimes. Returns are 100 times the log change in the real price index. * p < 0.10, ** p < 0.05, *** p < 0.01 (two-sided). Stars are shown only for tests against zero that are meaningful (means, AR term).

noisier, reacting to individual volatile months such as October 1987 and early 2020, and it is the only one a forecaster could have used.

5.2 Time-varying transitions fit better in sample

Table 4 lets the transition probabilities depend on lagged predictors. Inflation is the clearest single driver: higher inflation lowers the probability of staying in the bull regime (slope -1.28^{***} (0.32) on the logit scale), the likelihood ratio against fixed transitions is 15.5 ($p < 0.001$), and the regimes keep their meaning. The drawdown improves the likelihood more (35.9) but changes what the regimes describe: with it, the low-mean regime becomes short-lived (staying probability 0.214 at the average drawdown) and captures volatile months after a fall rather than whole bear markets. The dividend yield and volatility alone do not improve the fit significantly ($p = 0.228$ and 0.283). With all four predictors the likelihood ratio is 51.0 on 8 degrees of freedom and the AIC falls from 4,141.0 to 4,106.0, but the BIC, which penalizes the eight extra parameters more heavily, prefers fixed transitions (4,173.7 against 4,176.2); only 2 of the five TVTP models beat the fixed model on BIC. As with the drawdown alone, the joint model's low-mean regime is short-lived (mean −0.63% a month, staying probability 0.038 at the predictors' means), so its regimes are not the bull and bear regimes of the fixed model. The filtered bear probability of the joint model tracks the dated bear markets better in sample (AUC 0.865 against 0.802). Figure 3 shows why: the implied



Source: R. J. Shiller, U.S. Stock Markets 1871-Present and CAPE Ratio (shillerdata.com, downloaded 18 October 2026); author's calculations.

Figure 2: Filtered and smoothed bear-regime probabilities from the MS-AR(1) model, with dated bear markets shaded. Only the filtered probability is used for forecasting.

probability of leaving the bull regime swings between 0.005 and 1.000, against a constant 0.060 with fixed transitions, and it is high during and shortly after declines.

5.3 Inflation, valuation and weak returns are associated with bear markets

Table 5 reports full-sample binary models for the dated bear indicator, estimated on all origins from June 1881, and Figure 4 plots the logit coefficients. A higher dividend yield (0.98*** (0.37) at one month) and higher inflation (0.55*** (0.21)) raise the probability of a bear market at every horizon. A strong past year lowers the short-horizon probability (−0.68*** (0.22) at one month), which captures persistence: a low twelve-month return usually means a decline is already under way. Realized volatility matters mainly at long horizons (0.43*** (0.14) at twelve months). Because the dividend yield divides lagged dividends by the current price, it rises when prices fall, so part of its coefficient reflects an ongoing decline rather than cheapness; the dividend yield and the CAPE ratio are strongly negatively correlated, so their conditional coefficients should not be read separately. The pseudo-R^2 falls from 0.196 at one month to 0.073 at twelve months.

5.4 In real time, models beat the base rate; their edge over persistence is fragile

Table 6 is the central result. At the one-month horizon every model in the table beats the unconditional frequency (Brier 0.196). Persistence alone reaches 0.161 (skill 17.6%). The logit on the six main predictors reaches 0.146 (skill 25.4%, AUC 0.773), and the probit and the logit that adds the real-time state are almost identical. With the conventional Diebold-Mariano window these gains over persistence look significant ($p = 0.019$ for the logit), but with a window that covers

Table 4: Markov-switching models with time-varying transition probabilities

	Fixed transitions	Log dividend yield	CPI inflation, 12-month	12-month realized volatility	Drawdown from 12-month high	All four
Logit of P(bull to bull): slope on lagged predictor						
Log dividend yield		−0.60**				0.67
		(0.28)				(0.43)
CPI inflation, 12-month			−1.28***			−0.95**
			(0.32)			(0.40)
12-month realized volatility				−0.30		0.20
				(0.27)		(0.37)
Drawdown from 12-month high					2.39***	2.77**
					(0.87)	(1.26)
Logit of P(bear to bull): slope on lagged predictor						
Log dividend yield		0.80**				2.03
		(0.37)				(1.53)
CPI inflation, 12-month			0.56*			0.65
			(0.30)			(1.03)
12-month realized volatility				0.60*		0.71
				(0.35)		(0.92)
Drawdown from 12-month high					1.03	4.90
					(0.82)	(3.86)
Regimes (transition probabilities at the predictors' sample means)						
Bull mean / bear mean (%)	0.88 / −1.75	0.87 / −1.77	0.86 / −1.79	0.88 / −1.86	0.66 / −0.89	0.67 / −0.63
Bull sd / bear sd (%)	2.41 / 5.57	2.37 / 5.70	2.41 / 5.69	2.42 / 5.61	2.31 / 5.75	2.26 / 5.45
P(bull to bull)	0.940	0.915	0.926	0.943	0.738	0.657
P(bear to bear)	0.798	0.699	0.703	0.856	0.214	0.038
Fit						
Log-likelihood	−2,063.5	−2,062.0	−2,055.7	−2,062.2	−2,045.6	−2,038.0
LR statistic against fixed (df)		3.0 (2)	15.5 (2)	2.5 (2)	35.9 (2)	51.0 (8)
LR p-value		0.228	<0.001	0.283	<0.001	<0.001
AIC	4,141.0	4,142.0	4,129.5	4,142.5	4,109.1	4,106.0
BIC	4,173.7	4,184.2	4,171.6	4,184.6	4,151.2	4,176.2
AUC of filtered bear probability	0.802	0.812	0.874	0.807	0.861	0.865

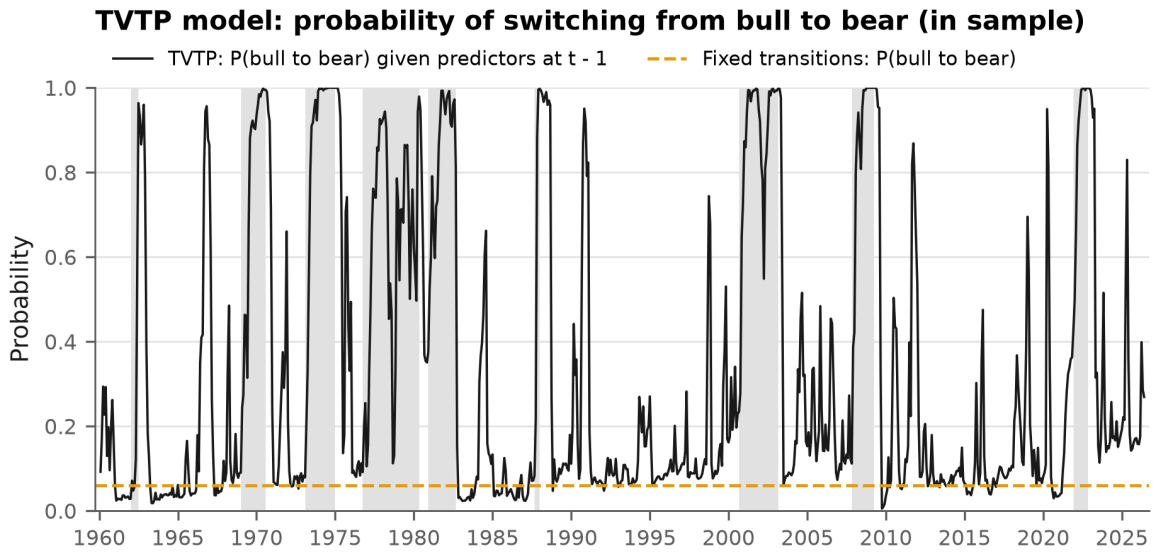
Notes: MS-AR(1) models of monthly real returns, 1960:02-2026:06. In the TVTP models the probability of moving from regime i at t - 1 to the bull regime at t is a logistic function of the standardized predictors at t - 1 (Diebold, Lee and Weinbach, 1994; Filardo, 1994). Slopes are per one standard deviation, with standard errors from the numerical Hessian in parentheses. The likelihood-ratio test compares each model with fixed transitions (chi-squared with two degrees of freedom per predictor). AUC compares the filtered bear-regime probability with the dated bear state (in sample). * p < 0.10, ** p < 0.05, *** p < 0.01 (two-sided).

the autocorrelation of the loss differences the p-value rises to 0.172. The MS-AR(1) forecast ranks months reasonably (AUC 0.713) but is less accurate than persistence (skill 9.1%). Against the unconditional frequency the window matters less at one month, where the number of models with a gain significant at the 5% level falls from 20 with the conventional window to 18 with 23 lags, than at three months, where it falls from 15 to 7.

The block bootstrap in Table 7 and Figure 5 puts these differences in perspective. The 90% interval for the one-month skill of persistence is [8.2, 26.5]% and that of the six-predictor logit [12.2, 39.4]%; both exclude zero. The logit's gain over persistence, however, ranges from −5.8 to 20.9 percentage points, and the logit beats persistence in a share of 0.84 of the bootstrap draws. With blocks of 12 or 36 months instead of 24 the interval for persistence becomes [9.4, 25.7]% or [8.7, 26.6]%.

5.5 Skill fades with the horizon

At three months the single-variable dividend-yield logit (skill 13.4%) and the six-predictor logit (12.4%) remain ahead of persistence (10.0%), but by margins that no test can distinguish from zero ($p = 0.338$ and 0.385 with the long window). At six months only the dividend-yield logit retains



Source: R. J. Shiller, U.S. Stock Markets 1871-Present and CAPE Ratio (shillerdata.com, downloaded 18 October 2026); author's calculations.

Figure 3: Probability of switching from the bull to the bear regime implied by the TVTP model with all four predictors (in sample, January 1960 to June 2026), against the constant probability of the fixed-transition model. Dated bear markets shaded.

a sizable gain (skill 8.2%, against 3.9% for persistence), and even that gain is not significant ($p = 0.245$ with the long window); the multivariate models are near zero. At twelve months no model is reliably better than the unconditional frequency: the best skill is 1.4% and the six-predictor logit has an AUC of 0.428, below the 0.5 of an uninformative forecast.

Table 5: Full-sample logit and probit models of a dated bear market h months ahead

	Logit				Probit			
	h = 1	h = 3	h = 6	h = 12	h = 1	h = 3	h = 6	h = 12
Valuation								
Log dividend yield	0.98***	0.99***	0.98***	1.01***	0.58***	0.59***	0.58***	0.60***
	(0.37)	(0.37)	(0.36)	(0.38)	(0.21)	(0.21)	(0.21)	(0.22)
Log CAPE	0.59*	0.65**	0.70**	0.77**	0.35*	0.39**	0.42**	0.47**
	(0.32)	(0.32)	(0.32)	(0.33)	(0.19)	(0.19)	(0.19)	(0.20)
Macroeconomic								
CPI inflation, 12-month	0.55***	0.52***	0.46**	0.41**	0.32***	0.31***	0.27**	0.25**
	(0.21)	(0.19)	(0.18)	(0.18)	(0.11)	(0.11)	(0.11)	(0.10)
12-month change in 10-year yield	0.25*	0.24*	0.25*	0.23*	0.15**	0.15*	0.15*	0.13*
	(0.13)	(0.13)	(0.13)	(0.13)	(0.08)	(0.08)	(0.08)	(0.08)
Market								
12-month real price change	−0.68***	−0.33*	−0.09	0.22	−0.38***	−0.20**	−0.07	0.13
	(0.22)	(0.19)	(0.16)	(0.16)	(0.11)	(0.10)	(0.09)	(0.10)
12-month realized volatility	0.17	0.24	0.30**	0.43***	0.11	0.15*	0.18**	0.26***
	(0.19)	(0.16)	(0.15)	(0.14)	(0.10)	(0.09)	(0.09)	(0.09)
Constant	−0.86***	−0.81***	−0.79***	−0.78***	−0.52***	−0.49***	−0.48***	−0.47***
	(0.16)	(0.16)	(0.16)	(0.17)	(0.09)	(0.09)	(0.09)	(0.10)
Pseudo-R^2 (McFadden)	0.196	0.130	0.089	0.073	0.196	0.130	0.089	0.073
Log-likelihood	−890.2	−961.8	−1,003.6	−1,014.9	−891.0	−961.4	−1,003.5	−1,015.7
Observations	1,740	1,738	1,735	1,729	1,740	1,738	1,735	1,729

Notes: Dependent variable: 1 if month t+h lies in a dated bear market. Predictors are measured at t and standardized. Newey-West standard errors in parentheses (10, 10, 10, 12 lags by horizon). Forecast origins 1881:06 to 2026:05 for h = 1. * p < 0.10, ** p < 0.05, *** p < 0.01 (two-sided). In-sample estimates over the whole period, not real-time forecasts.

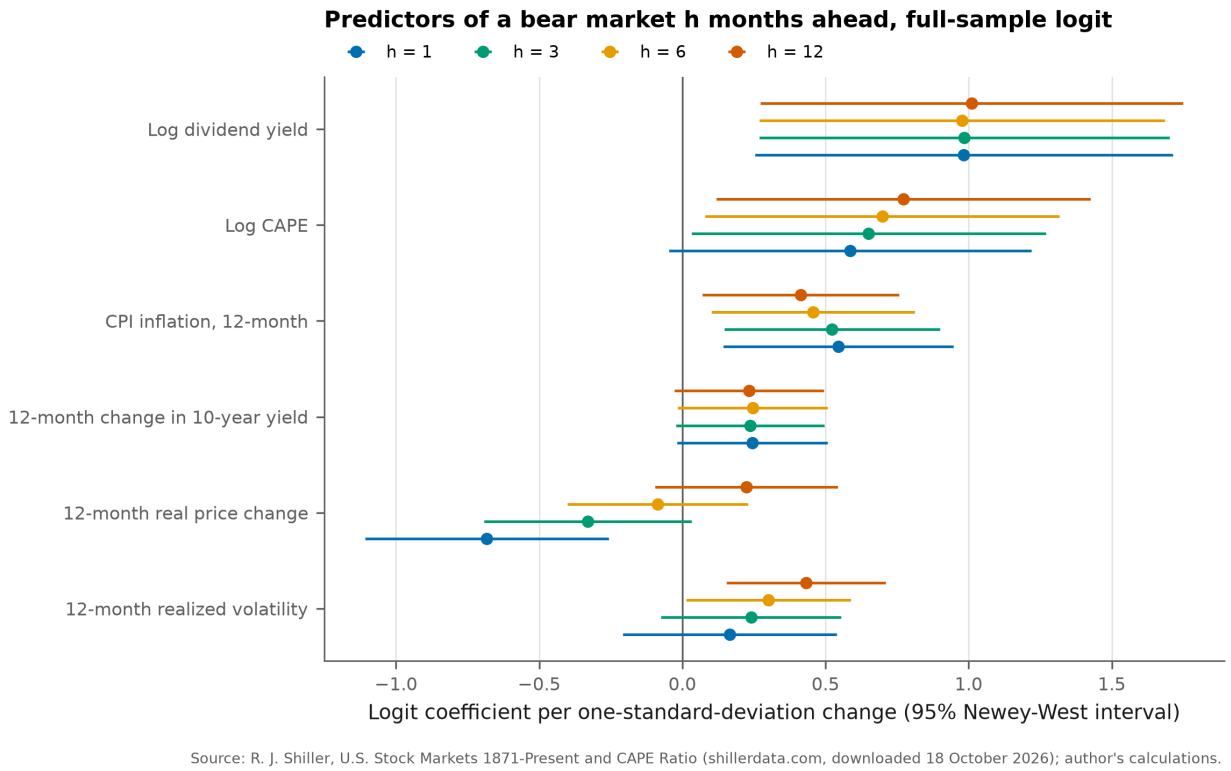


Figure 4: Full-sample logit coefficients per one-standard-deviation change in each predictor, with 95% Newey-West confidence intervals, by forecast horizon.

Figure 6 shows why. The model probabilities rise sharply once a decline is under way and fall when the market recovers, so at short horizons they track bear markets well, mostly with a lag. The calibration curves in Figure 7 show that the short-horizon forecasts are directionally right, while at twelve months the curves are flat, and the ROC curves in Figure 8 lie close to the diagonal.

5.6 Time-varying transitions, machine learning and combinations

Table 8 evaluates the newer methods in the same real-time design. The TVTP model ranks months better than the fixed-transition model (one-month AUC 0.769 against 0.713) but its probabilities are worse calibrated, and its Brier skill (8.0%) is no higher than that of the fixed model (9.1%) and below persistence at every horizon. Its in-sample advantage does not carry over to forecasting.

The machine-learning models do not change the picture either. Gradient boosting on the rich predictor set is the most accurate of them at one month (skill 26.8%, AUC 0.793), essentially tied with the six-predictor logit, but it deteriorates sharply as the horizon lengthens (−12.3% at six months and −32.1% at twelve): with only a handful of bear markets in each training window, flexible models fit episodes rather than regularities. The elastic net is less accurate than the unpenalized six-predictor logit (19.5% against 25.4% at one month); part of this gap may reflect the annual rather than monthly re-estimation of the machine-learning models, which updates their training labels only once a year. The time-series validation most often chose the weakest penalty for the elastic net on the main predictors (C = 10 and mixing weight 0.1, 39% of the 266 fits) and the shallowest, shortest boosting schedule for both predictor sets (depth 2 and 100 rounds, 80% of fits on the rich set).

Table 6: Out-of-sample accuracy of real-time bear-market forecasts, targets 1960:02-2026:06

	Brier	Skill (%)	Log loss	AUC	p vs uncond.	p vs persist.	p vs persist., 23 lags
h = 1 (797 forecasts, bear share 21.8%)							
Unconditional frequency	0.196	0.0	0.583	0.585			
Persistence (real-time state)	0.161	17.6	0.506	0.707	<0.001		
MS-AR(1) filtered probability	0.178	9.1	0.542	0.713	<0.001	0.999	0.938
Logit: dividend yield	0.160	18.3	0.509	0.646	<0.001	0.441	0.475
Logit: 6 main predictors	0.146	25.4	0.460	0.773	<0.001	0.019	0.172
Probit: 6 main predictors	0.146	25.3	0.463	0.772	<0.001	0.021	0.177
Logit: 6 predictors + state	0.148	24.6	0.464	0.772	<0.001	0.022	0.171
h = 3 (795 forecasts, bear share 21.9%)							
Unconditional frequency	0.196	0.0	0.584	0.569			
Persistence (real-time state)	0.177	10.0	0.540	0.651	<0.001		
MS-AR(1) filtered probability	0.191	2.7	0.570	0.635	0.137	0.999	0.967
Logit: dividend yield	0.170	13.4	0.529	0.607	<0.001	0.202	0.338
Logit: 6 main predictors	0.172	12.4	0.528	0.671	0.005	0.290	0.385
Probit: 6 main predictors	0.172	12.4	0.531	0.672	0.006	0.295	0.388
Logit: 6 predictors + state	0.174	11.2	0.534	0.667	0.013	0.383	0.436
h = 6 (792 forecasts, bear share 22.0%)							
Unconditional frequency	0.197	0.0	0.584	0.550			
Persistence (real-time state)	0.189	3.9	0.567	0.586	0.028		
MS-AR(1) filtered probability	0.199	−1.4	0.589	0.556	0.696	0.989	0.965
Logit: dividend yield	0.180	8.2	0.552	0.558	0.020	0.146	0.245
Logit: 6 main predictors	0.193	1.7	0.577	0.550	0.388	0.660	0.606
Probit: 6 main predictors	0.194	1.3	0.581	0.549	0.412	0.682	0.621
Logit: 6 predictors + state	0.196	0.1	0.584	0.544	0.495	0.761	0.680
h = 12 (786 forecasts, bear share 22.1%)							
Unconditional frequency	0.197	0.0	0.584	0.535			
Persistence (real-time state)	0.197	−0.2	0.585	0.498	0.625		
MS-AR(1) filtered probability	0.200	−1.8	0.591	0.496	0.827	0.820	0.797
Logit: dividend yield	0.194	1.4	0.579	0.489	0.353	0.334	0.360
Logit: 6 main predictors	0.203	−3.2	0.605	0.428	0.679	0.673	0.645
Probit: 6 main predictors	0.205	−4.3	0.611	0.419	0.726	0.721	0.686
Logit: 6 predictors + state	0.204	−3.8	0.609	0.423	0.706	0.700	0.669

Notes: Expanding-window forecasts made at each month-end origin using only data available then: predictors with publication lags, bear dates obtained from prices through the origin, and models re-estimated on that information. Lower Brier score and log loss are better; skill is the percentage reduction in Brier score relative to the unconditional frequency. Log loss truncates probabilities to [0.001, 0.999]. The p-values are one-sided Diebold-Mariano tests (Harvey-Leybourne-Newbold correction) of equal Brier score against the alternative that the model is more accurate than the baseline, with the conventional max(h - 1, 1) Bartlett lags; the last column uses 23 lags, which allows for the strong autocorrelation of loss differences during long bear markets. Markov-switching parameters are re-estimated each January (filtered probabilities only).

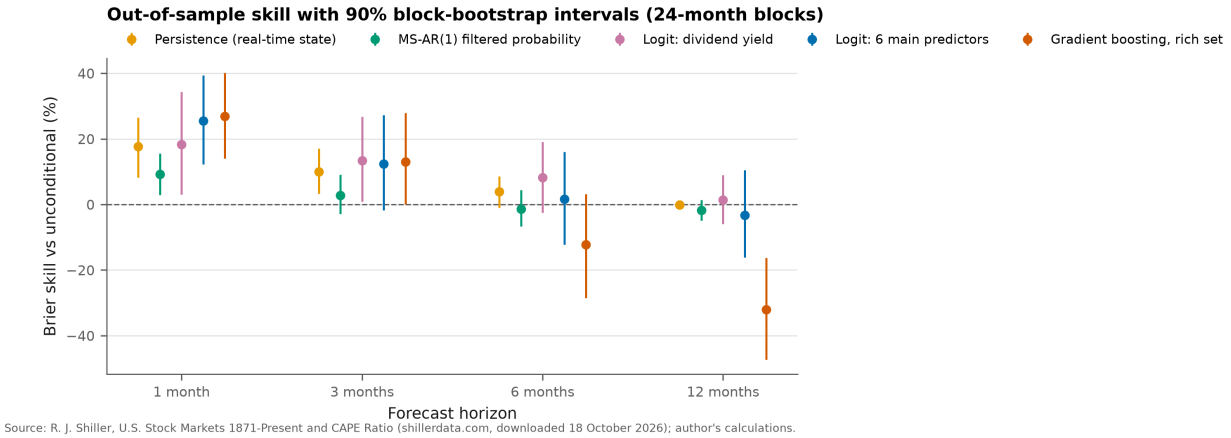


Figure 5: Out-of-sample Brier skill relative to the unconditional frequency, with 90% moving-block bootstrap intervals. The dashed line is the unconditional frequency.

Table 7: Block-bootstrap uncertainty of out-of-sample Brier skill

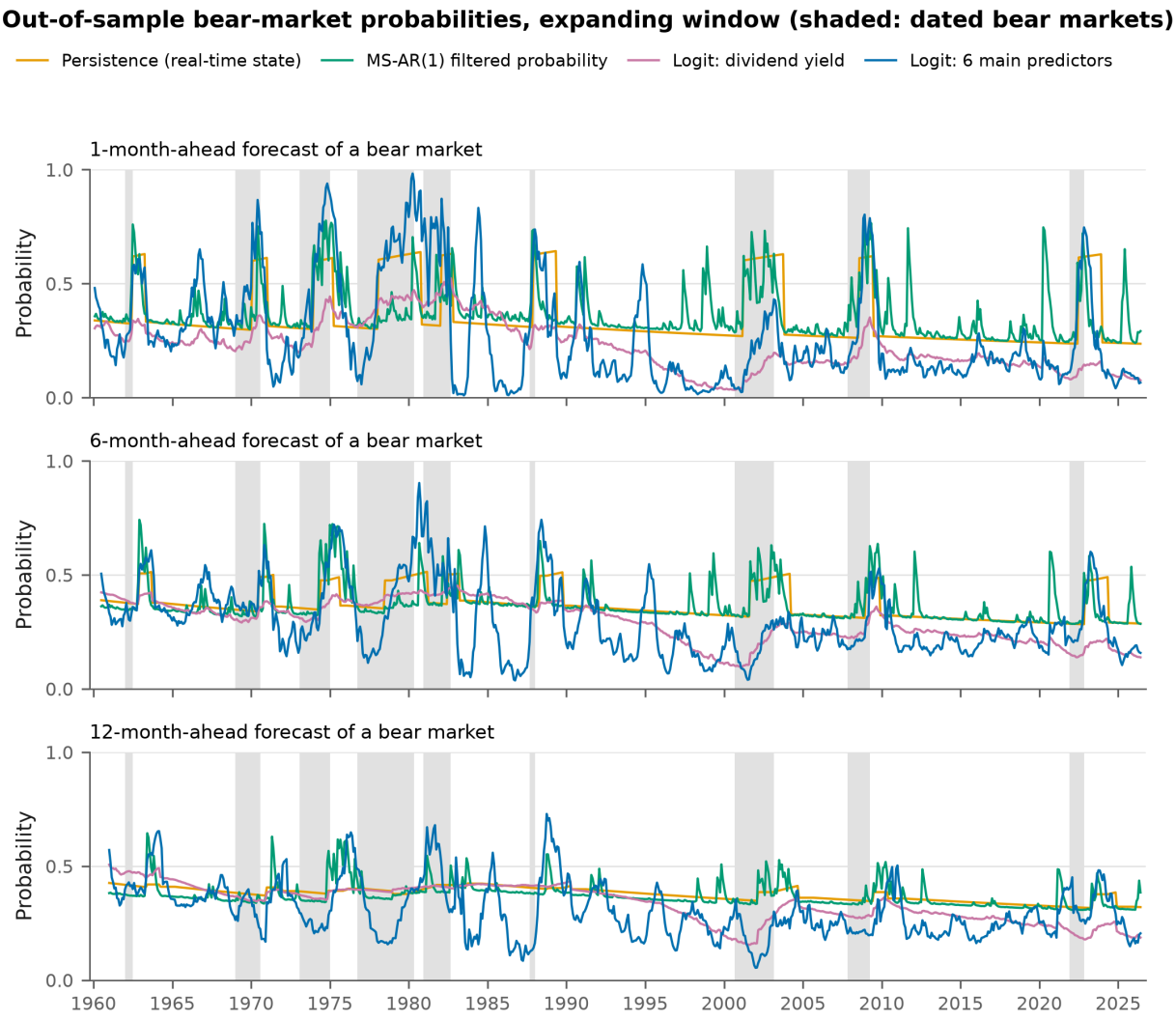
	Skill (%)	90% interval	Gain over persistence (pp), 90% interval	Share of draws beating persistence
h = 1				
Persistence (real-time state)	17.6	[8.2, 26.5]		
MS-AR(1) filtered probability	9.1	[3.0, 15.6]	[−17.7, 1.0]	0.07
TVTP MS-AR(1)	8.0	[−1.4, 17.0]	[−19.5, −0.3]	0.04
Logit: dividend yield	18.3	[3.0, 34.4]	[−15.9, 17.3]	0.57
Logit: 6 main predictors	25.4	[12.2, 39.4]	[−5.8, 20.9]	0.84
Elastic-net logit, rich set	17.3	[2.6, 31.5]	[−15.2, 13.7]	0.52
Gradient boosting, rich set	26.8	[14.0, 40.1]	[−4.5, 22.9]	0.87
DMA: all models	23.4	[11.6, 35.5]	[−3.4, 15.5]	0.86
h = 3				
Persistence (real-time state)	10.0	[3.3, 17.1]		
MS-AR(1) filtered probability	2.7	[−2.9, 9.0]	[−13.9, −0.8]	0.03
TVTP MS-AR(1)	2.8	[−5.7, 11.0]	[−15.7, 0.0]	0.05
Logit: dividend yield	13.4	[0.9, 26.7]	[−9.8, 16.5]	0.69
Logit: 6 main predictors	12.4	[−1.8, 27.2]	[−10.6, 15.7]	0.64
Elastic-net logit, rich set	3.9	[−10.4, 18.2]	[−19.7, 7.7]	0.24
Gradient boosting, rich set	13.0	[0.0, 27.9]	[−10.0, 16.6]	0.65
DMA: all models	9.4	[−2.3, 22.2]	[−9.8, 9.0]	0.48
h = 6				
Persistence (real-time state)	3.9	[−1.0, 8.6]		
MS-AR(1) filtered probability	−1.4	[−6.7, 4.4]	[−10.0, −0.9]	0.03
TVTP MS-AR(1)	−1.3	[−8.4, 5.6]	[−11.2, 0.5]	0.07
Logit: dividend yield	8.2	[−2.5, 19.1]	[−6.3, 14.2]	0.76
Logit: 6 main predictors	1.7	[−12.2, 16.1]	[−15.1, 10.9]	0.41
Elastic-net logit, rich set	−1.4	[−13.6, 10.0]	[−17.3, 5.8]	0.24
Gradient boosting, rich set	−12.3	[−28.5, 3.2]	[−31.7, −1.8]	0.03
DMA: all models	−0.9	[−10.7, 9.2]	[−12.9, 3.2]	0.16
h = 12				
Persistence (real-time state)	−0.2	[−1.0, 0.7]		
MS-AR(1) filtered probability	−1.8	[−5.0, 1.4]	[−4.7, 1.4]	0.18
TVTP MS-AR(1)	−3.9	[−9.0, 1.2]	[−8.7, 1.0]	0.09
Logit: dividend yield	1.4	[−5.9, 9.0]	[−5.6, 9.0]	0.66
Logit: 6 main predictors	−3.2	[−16.1, 10.5]	[−15.8, 10.4]	0.36
Elastic-net logit, rich set	−8.6	[−20.5, 2.5]	[−20.5, 2.5]	0.11
Gradient boosting, rich set	−32.1	[−47.3, −16.3]	[−47.1, −16.2]	0.00
DMA: all models	−4.5	[−13.9, 4.7]	[−13.7, 4.5]	0.24

Notes: Circular moving-block bootstrap of target months with blocks of 24 months and 2,000 replications, resampling all models jointly. Intervals are 5th and 95th percentiles. The gain over persistence is the model's skill minus the skill of persistence in the same draw.

Combinations are stable rather than spectacular (Table 9). Dynamic model averaging of the main models reaches a one-month skill of 23.4%, below the best individual model in hindsight (29.1% for the logit on all 11 predictors) but without having to know in advance which model that is; it inherits the weakness of its members beyond three months (−4.5% at twelve). The equal-weight average of the univariate logits is more conservative (13.0% at one month, 2.7% at six). The forgetting factor is not innocuous: with α between 0.95 and 1 the one-month skill of DMA on all models ranges from 22.6% to 27.3%, a spread comparable to the gain of the best models over persistence, and for DMA on all models the fastest forgetting is the most accurate at every horizon. Because that ranking is only known in hindsight, the main results keep $\alpha = 0.99$.

5.7 Where the skill comes from

The Murphy decomposition in Table 10 and Figure 9 separates calibration from discrimination. At one month persistence already has about as much resolution as the six-predictor logit (3.22 against 3.09, times 100) but is worse calibrated (reliability 2.13 against 0.81). The logit's advantage over



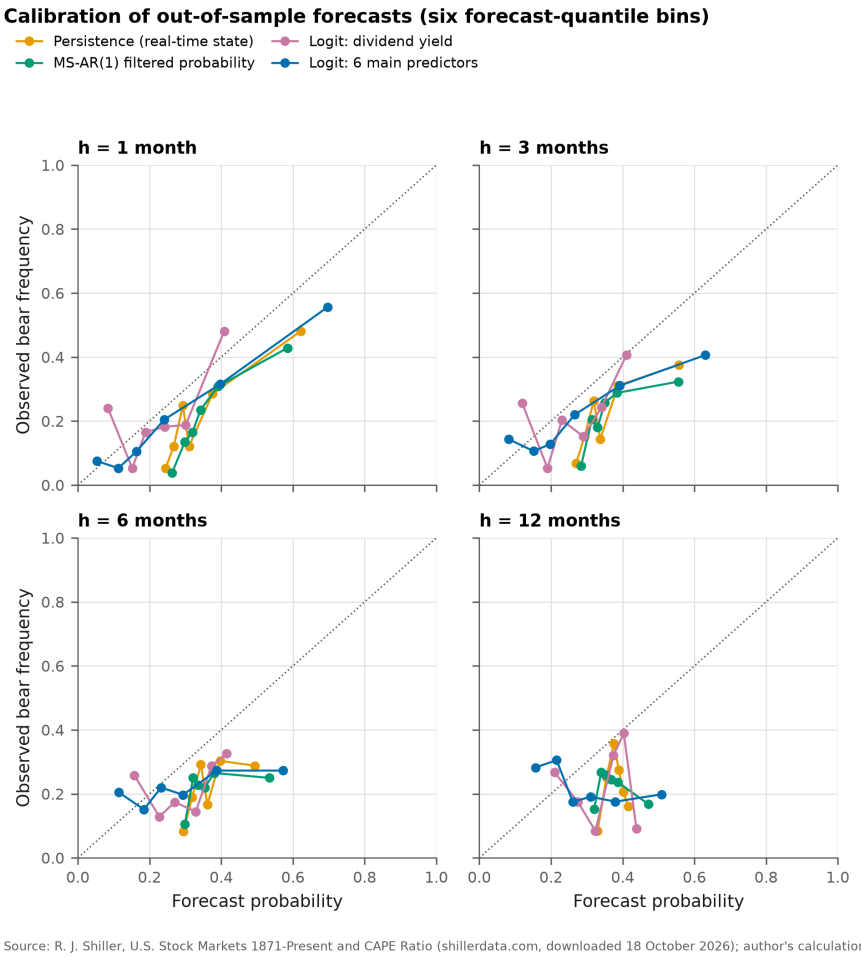
Source: R. J. Shiller, U.S. Stock Markets 1871-Present and CAPE Ratio (shillerdata.com, downloaded 18 October 2026); author's calculations.

Figure 6: Real-time forecast probabilities, plotted at the target month, with dated bear markets shaded. Each forecast uses only data available at its origin.

persistence is therefore mostly better calibration of the same information, not new information. The univariate drawdown logit has the highest resolution at one month, which is consistent with its role as an early measure of a decline already in progress. At twelve months resolution is small for the main models (0.42 for the logit and 0.08 for persistence), and the forecasts with more resolution, such as boosting on the rich set, pay for it with much worse reliability (0.99 against 9.35).

5.8 Accounting for the number of models compared

The main design compares 24 distinct non-baseline forecasts with persistence at four horizons, 96 comparisons in all (the univariate dividend-yield logit is the dividend-yield logit of Table 6 and is counted once). With the conventional Diebold-Mariano window, 9 of them are significant at the 5% level (9 at one month); with the 23-lag window 1 remains (Univariate logit: Drawdown from 12-month high, $p = 0.033$). The tests in Table 11 and Table 12 settle the question formally. Hansen's SPA test cannot reject that no model beats persistence: the consistent p-value is 0.189 at one



Source: R. J. Shiller, U.S. Stock Markets 1871-Present and CAPE Ratio (shillerdata.com, downloaded 18 October 2026); author's calculations.

Figure 7: Calibration of out-of-sample forecasts: observed bear frequency against the mean forecast within six forecast-quantile bins. The dotted line is perfect calibration.

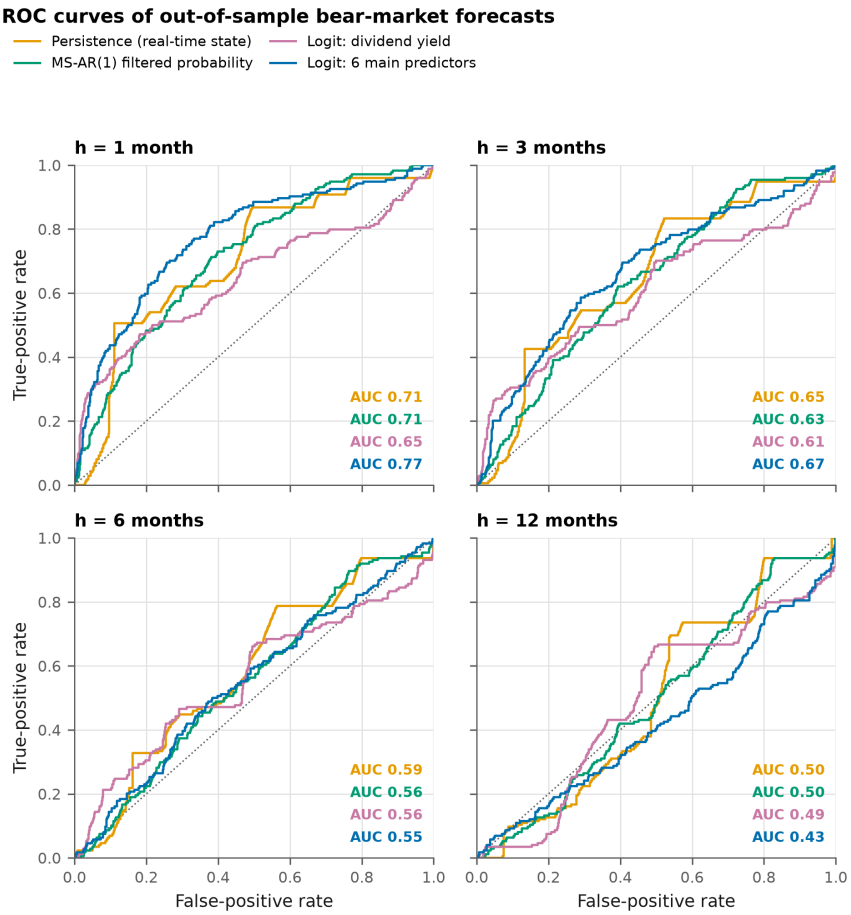
month, 0.532 at three, and 0.468 when all 96 model-horizon pairs are considered jointly (Reality Check 0.502; 0.446 with 12-month blocks). Persistence belongs to the 90% model confidence set at every horizon (MCS p-values 0.788, 0.990, 0.992 and 0.997), whereas the unconditional frequency is excluded at one month ($p = 0.002$). At one month 18 of the 26 models are in the set; at longer horizons almost all are, because the data cannot rank them.

The Giacomini-White test asks whether the advantage of the most accurate model at each horizon depends on whether a bear market is under way. At one month it does not ($p = 0.325$): the best model (Logit: all 11 predictors) is ahead of persistence by a similar margin in both states. At three months there is weak evidence ($p = 0.086$) that the drawdown logit helps in bull states (mean loss difference -10.7, times 1,000) and hurts in confirmed bear markets (5.5), which fits its role as an early-warning measure of declines in progress.

6 Robustness

6.1 Regime specification and sample

Table 13 compares the main model with four alternatives on January 1960 to June 2026 and with the main model estimated on two subsamples. The MS-AR(1) has the lowest AIC (4,141.0) and



Source: R. J. Shiller, U.S. Stock Markets 1871-Present and CAPE Ratio (shillerdata.com, downloaded 18 October 2026); author's calculations.

Figure 8: ROC curves of out-of-sample forecasts by horizon, with areas under the curve.

BIC (4,173.7) in the first group. Letting only the variance switch gives a high-volatility regime (standard deviation 6.12% against 2.54%) that tracks dated bear markets less well (AUC 0.747 against 0.802). Letting only the mean switch picks out crash months rather than bear markets: the low-mean regime averages −9.81% a month and lasts 1.4 months. A three-regime model splits the bull state into a calm regime (mean 0.72%, duration 17.1 months) and a short rebound regime (mean 3.58%, duration 3.1 months), while its bear regime (mean −2.63%, standard deviation 4.91%) resembles the two-regime one and tracks dated bear markets better (AUC 0.879); its extra parameters do not pay for themselves by AIC (4,152.2). Across subsamples the bear regime's volatility is stable (5.63% before 1990, 5.63% after), whereas the bull regime became more persistent (expected duration 10.8 months before, 17.7 after) and the filtered probability became a much better real-time marker of bear markets (AUC 0.732 before, 0.900 after).

6.2 Dating rule

Table 14 shows how the set of episodes depends on the rule. A 15% threshold dates 38 bear markets since 1871 and a 25% threshold 20, against 25 for the main rule. Dating nominal prices yields 7 episodes since 1960, because inflation in the 1970s turned long real declines into flat nominal prices; the bear share falls to 13.5%. Table 15 re-runs the entire real-time exercise under each rule. The six-predictor logit has a one-month skill between 25.1% and 26.4% under every rule,

Table 8: Out-of-sample accuracy of time-varying-transition, machine-learning and combination forecasts, targets 1960:02-2026:06

	Brier	Skill (%)	Log loss	AUC	p vs uncond.	p vs persist.	p vs persist., 23 lags
h = 1 (797 forecasts, bear share 21.8%)							
Unconditional frequency	0.196	0.0	0.583	0.585			
Persistence (real-time state)	0.161	17.6	0.506	0.707	<0.001		
TVTP MS-AR(1)	0.180	8.0	0.546	0.769	0.005	1.000	0.956
Elastic-net logit	0.158	19.5	0.498	0.738	<0.001	0.311	0.416
Elastic-net logit, rich set	0.162	17.3	0.529	0.729	<0.001	0.532	0.514
Gradient boosting	0.159	18.8	0.494	0.752	<0.001	0.401	0.451
Gradient boosting, rich set	0.143	26.8	0.444	0.793	<0.001	0.015	0.136
Equal-weight combination	0.171	13.0	0.527	0.783	<0.001	0.972	0.824
DMA: univariate logits	0.150	23.3	0.485	0.772	<0.001	0.022	0.101
DMA: all models	0.150	23.4	0.471	0.757	<0.001	0.032	0.153
h = 3 (795 forecasts, bear share 21.9%)							
Unconditional frequency	0.196	0.0	0.584	0.569			
Persistence (real-time state)	0.177	10.0	0.540	0.651	<0.001		
TVTP MS-AR(1)	0.191	2.8	0.569	0.712	0.191	0.995	0.942
Elastic-net logit	0.181	7.7	0.555	0.661	0.027	0.719	0.613
Elastic-net logit, rich set	0.189	3.9	0.584	0.643	0.204	0.916	0.754
Gradient boosting	0.192	2.1	0.591	0.645	0.364	0.927	0.792
Gradient boosting, rich set	0.171	13.0	0.515	0.697	0.006	0.272	0.362
Equal-weight combination	0.182	7.2	0.552	0.693	<0.001	0.911	0.773
DMA: univariate logits	0.175	10.6	0.538	0.696	0.004	0.422	0.447
DMA: all models	0.178	9.4	0.540	0.658	0.019	0.567	0.539
h = 6 (792 forecasts, bear share 22.0%)							
Unconditional frequency	0.197	0.0	0.584	0.550			
Persistence (real-time state)	0.189	3.9	0.567	0.586	0.028		
TVTP MS-AR(1)	0.199	−1.3	0.587	0.652	0.647	0.965	0.931
Elastic-net logit	0.191	2.7	0.577	0.586	0.278	0.600	0.565
Elastic-net logit, rich set	0.199	−1.4	0.602	0.541	0.611	0.866	0.763
Gradient boosting	0.206	−4.7	0.615	0.538	0.759	0.913	0.822
Gradient boosting, rich set	0.221	−12.3	0.657	0.530	0.945	0.984	0.948
Equal-weight combination	0.191	2.7	0.572	0.587	0.007	0.761	0.692
DMA: univariate logits	0.192	2.3	0.574	0.610	0.301	0.674	0.645
DMA: all models	0.198	−0.9	0.592	0.546	0.577	0.896	0.821
h = 12 (786 forecasts, bear share 22.1%)							
Unconditional frequency	0.197	0.0	0.584	0.535			
Persistence (real-time state)	0.197	−0.2	0.585	0.498	0.625		
TVTP MS-AR(1)	0.204	−3.9	0.598	0.573	0.901	0.902	0.887
Elastic-net logit	0.201	−2.4	0.603	0.502	0.679	0.668	0.644
Elastic-net logit, rich set	0.214	−8.6	0.634	0.466	0.917	0.914	0.874
Gradient boosting	0.226	−15.1	0.668	0.403	0.959	0.959	0.933
Gradient boosting, rich set	0.260	−32.1	0.761	0.351	1.000	1.000	0.999
Equal-weight combination	0.198	−0.7	0.587	0.472	0.789	0.742	0.714
DMA: univariate logits	0.205	−4.5	0.605	0.419	0.909	0.912	0.885
DMA: all models	0.205	−4.5	0.610	0.467	0.826	0.822	0.771

Notes: Expanding-window forecasts made at each month-end origin using only data available then: predictors with publication lags, bear dates obtained from prices through the origin, and models re-estimated on that information. Lower Brier score and log loss are better; skill is the percentage reduction in Brier score relative to the unconditional frequency. Log loss truncates probabilities to [0.001, 0.999]. The p-values are one-sided Diebold-Mariano tests (Harvey-Leybourne-Newbold correction) of equal Brier score against the alternative that the model is more accurate than the baseline, with the conventional max(h - 1, 1) Bartlett lags; the last column uses 23 lags, which allows for the strong autocorrelation of loss differences during long bear markets. The TVTP model and the machine-learning models are re-estimated each January; the elastic net and gradient boosting choose their hyperparameters by validation on the last 10 years of each training window. The rich set adds the five additional predictors and the real-time bear state.

persistence between 10.1% (25%) and 22.9% (15%), and the multivariate logit has negative skill at twelve months under both real-price rules (−5.8% and −7.2%). The SPA test against persistence does not reject under any alternative rule or window (smallest consistent p-value 0.242, Table 11).

Table 9: Forecast combinations: out-of-sample Brier skill (%)

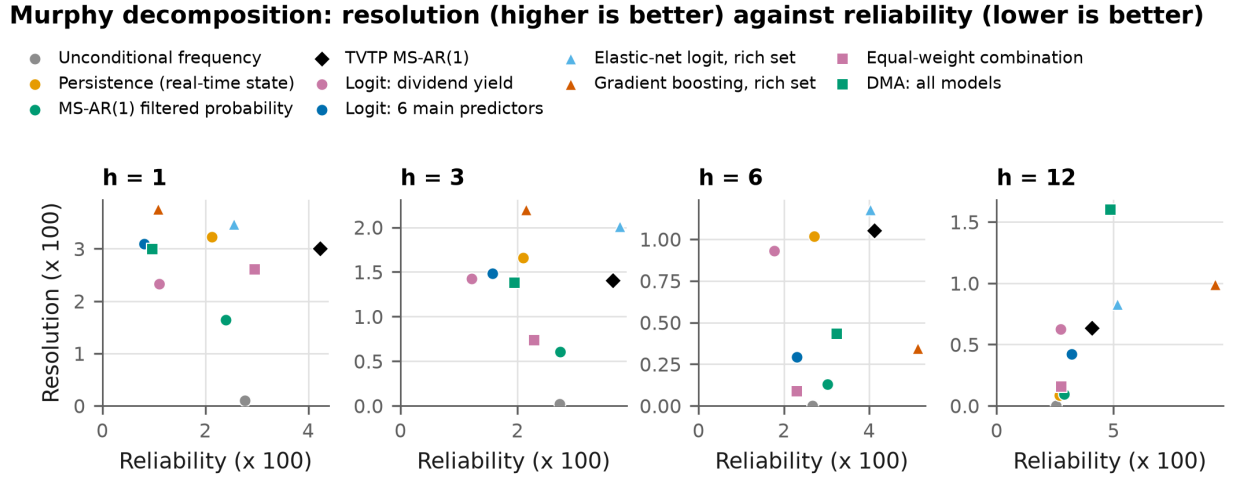
	h = 1	h = 3	h = 6	h = 12
Equal-weight combination	13.0***	7.2***	2.7***	−0.7
DMA, univariate logits, alpha = 0.95	24.0***	10.5***	0.6	−4.8
DMA, all models, alpha = 0.95	27.3***	13.2***	3.1	−2.2
DMA, univariate logits, alpha = 0.99	23.3***	10.6***	2.3	−4.5
DMA, all models, alpha = 0.99	23.4***	9.4**	−0.9	−4.5
DMA, univariate logits, alpha = 1	22.1***	7.8**	−0.9	−6.3
DMA, all models, alpha = 1	22.6***	6.8*	−0.4	−5.5

Notes: The equal-weight combination averages the univariate logits. Dynamic model averaging (Raftery, Kárný and Ettler, 2010) weights forecasts by their exponentially discounted predictive likelihood, using only outcomes known at the origin; alpha is the forgetting factor (alpha = 1 is recursive Bayesian model averaging). Stars: one-sided Diebold-Mariano test against the unconditional frequency, * p < 0.10, ** p < 0.05, *** p < 0.01.

Table 10: Murphy decomposition of out-of-sample Brier scores (x 100)

	h = 1		h = 3		h = 6		h = 12	
	Rel.	Res.	Rel.	Res.	Rel.	Res.	Rel.	Res.
Unconditional frequency	2.77	0.10	2.72	0.02	2.67	0.00	2.54	0.00
Persistence (real-time state)	2.13	3.22	2.10	1.66	2.71	1.02	2.68	0.08
MS-AR(1) filtered probability	2.40	1.64	2.73	0.61	3.03	0.13	2.89	0.10
TVTP MS-AR(1)	4.23	3.00	3.63	1.41	4.12	1.05	4.07	0.64
Logit: dividend yield	1.10	2.33	1.21	1.42	1.77	0.93	2.74	0.62
Logit: 6 main predictors	0.81	3.09	1.57	1.48	2.30	0.29	3.21	0.42
Logit: 6 predictors + state	0.91	3.14	1.56	1.25	2.85	0.43	3.67	0.81
Elastic-net logit	1.46	2.83	3.60	2.67	3.74	1.70	4.22	1.26
Elastic-net logit, rich set	2.56	3.47	3.75	2.01	4.02	1.18	5.15	0.83
Gradient boosting	2.32	3.52	3.53	1.50	3.88	0.58	5.71	0.51
Gradient boosting, rich set	1.08	3.76	2.15	2.20	5.14	0.34	9.35	0.99
Equal-weight combination	2.96	2.60	2.29	0.73	2.31	0.09	2.76	0.16
DMA: all models	0.97	2.99	1.94	1.38	3.24	0.43	4.86	1.60
Uncertainty (all models)	17.07		17.10		17.14		17.24	

Notes: Brier = reliability - resolution + uncertainty (Murphy, 1973), with forecasts grouped into 10 equal-width bins. Reliability (lower is better) measures miscalibration; resolution (higher is better) measures how much the observed bear frequency varies across forecast bins. The remainder due to variation of forecasts within bins is at most 0.44 (x 100) in absolute value.



Source: R. J. Shiller, U.S. Stock Markets 1871-Present and CAPE Ratio (shillerdata.com, downloaded 18 October 2026); author's calculations.

Figure 9: Murphy decomposition of out-of-sample Brier scores: resolution against reliability for selected models, by horizon.

Table 11: Tests of superior and conditional predictive ability against persistence

	Models	Best model	p-value	Reality Check p / GW stat (DM p)
Superior predictive ability against persistence				
Main design, h = 1	24	Univariate logit: Drawdown from 12-month high	0.189	0.205
Main design, h = 3	24	Univariate logit: Drawdown from 12-month high	0.532	0.557
Main design, h = 6	24	Logit: dividend yield	0.830	0.859
Main design, h = 12	24	Univariate logit: CPI inflation, 12-month	0.977	0.988
Main design, all horizons	96	Univariate logit: Drawdown from 12-month high	0.468	0.502
Dating threshold 15%, all horizons	12	Logit: 6 main predictors	0.880	0.940
Dating threshold 25%, all horizons	12	Logit: 6 main predictors	0.242	0.242
Nominal prices, 20%, all horizons	12	Logit: 6 main predictors	0.323	0.323
Rolling 30-year window, all horizons	12	Logit: 6 main predictors	0.681	0.722
Conditional predictive ability: best model against persistence				
h = 1		Logit: all 11 predictors	0.325	2.25 (0.081)
h = 3		Univariate logit: Drawdown from 12-month high	0.086	4.91 (0.131)
h = 6		Logit: dividend yield	0.624	0.94 (0.245)
h = 12		Logit: dividend yield	0.934	0.14 (0.360)

Notes: Upper panel: Hansen (2005) consistent SPA p-value and White (2000) Reality Check p-value for the null that no model has lower expected Brier loss than persistence, circular block bootstrap with 24-month blocks and 2,000 draws; 'Best model' has the largest studentized gain. The joint rows treat every (model, horizon) pair as a separate forecast on the origins common to all horizons. Lower panel: Giacomini and White (2006) Wald test of equal conditional accuracy of the most accurate model and persistence, with test functions (1, real-time bear state) known at the origin and a Bartlett long-run variance with 23 lags; the last column gives the statistic and, in parentheses, the one-sided unconditional Diebold-Mariano p-value with the same 23 lags.

6.3 Estimation window and evaluation period

Rolling 30-year windows reduce the skill of persistence, the regime model and the dividend-yield logit at every horizon (for example 3.0% for the dividend-yield logit at one month, against 18.3%), and the multivariate logit deteriorates sharply at long horizons (−54.7% at twelve months). The longer history in the expanding window is valuable, which is expected for a rare event: thirty years contain only a handful of bear markets.

The evaluation-period rows of Table 15 show that forecastability is concentrated in the later part of the sample. For targets from 1990 onward, persistence reaches a one-month skill of 28.3% and the multivariate logit 34.0%; for earlier targets the corresponding figures are 8.0% and 17.7%.

Table 12: Model confidence set p-values for out-of-sample Brier loss

	h = 1	h = 3	h = 6	h = 12
Unconditional frequency	0.002	0.370*	0.981*	0.997*
Persistence (real-time state)	0.788*	0.990*	0.992*	0.997*
MS-AR(1) filtered probability	0.204*	0.684*	0.972*	0.993*
TVTP MS-AR(1)	0.234*	0.750*	0.981*	0.947*
Logit: dividend yield	0.806*	1.000*	1.000*	1.000*
Logit: 6 main predictors	0.940*	1.000*	0.992*	0.997*
Probit: 6 main predictors	0.940*	1.000*	0.992*	0.976*
Logit: 6 predictors + state	0.940*	0.990*	0.981*	0.993*
Elastic-net logit	0.788*	0.781*	0.992*	0.997*
Elastic-net logit, rich set	0.546*	0.722*	0.981*	0.837*
Gradient boosting	0.806*	0.750*	0.968*	0.638*
Gradient boosting, rich set	0.959*	1.000*	0.464*	0.021
Logit: all 11 predictors	1.000*	1.000*	0.981*	0.646*
Equal-weight combination	0.124*	0.781*	0.992*	0.997*
DMA: univariate logits	0.940*	0.990*	0.992*	0.837*
DMA: all models	0.908*	0.860*	0.981*	0.947*

Notes: Hansen, Lunde and Nason (2011) T_max procedure on squared forecast errors, circular block bootstrap with 24-month blocks and 2,000 draws. A star marks models in the 90% model confidence set (p >= 0.10): the set of models that cannot be distinguished from the best at the 10% level.

Table 13: Robustness of the regime model: specification and sample

	MS-AR(1)	Mean and variance	Variance only	Mean only	Three regimes	MS-AR(1), 1960-1989	MS-AR(1), 1990-2026
Mean monthly real return (%)							
Bull (highest mean)	0.88	0.95	0.65	0.71	3.58	0.62	1.10
Middle (three regimes)					0.72		
Bear (lowest mean)	−1.75	−1.84	0.65	−9.81	−2.63	−2.45	−1.48
Volatility (sd, %)							
Bull (highest mean)	2.41	2.49	2.54	3.01	2.71	2.57	2.24
Middle (three regimes)					2.33		
Bear (lowest mean)	5.57	5.51	6.12	3.01	4.91	5.63	5.63
Expected duration (months)							
Bull (highest mean)	16.6	18.1	21.4	35.2	3.1	10.8	17.7
Middle (three regimes)					17.1		
Bear (lowest mean)	5.0	5.5	6.1	1.4	5.8	2.4	5.9
Fit							
AR(1) coefficient	0.196					0.242	0.143
Parameters	7	6	5	5	12	7	7
Log-likelihood	−2,063.5	−2,081.2	−2,094.8	−2,104.1	−2,064.1	−935.9	−1,119.9
AIC	4,141.0	4,174.3	4,199.6	4,218.1	4,152.2	1,885.8	2,253.8
BIC	4,173.7	4,202.4	4,223.0	4,241.5	4,208.4	1,913.0	2,282.4
Observations	797	798	798	798	798	359	437
Filtered bear probability against the dated bear state							
AUC	0.802	0.830	0.747	0.771	0.879	0.732	0.900
Correlation	0.42	0.46	0.35	0.30	0.56	0.31	0.56

Notes: Two-regime models unless stated; monthly real returns. The first five columns use January 1960 to June 2026; AIC and BIC are comparable only within that group. 'Mean and variance' is the switching mean-variance model without the AR term; 'Variance only' lets only the variance switch (the bear regime is the high-variance one); 'Mean only' lets only the mean switch. In the three-regime model the bear regime has the lowest mean. The last two columns re-estimate the main MS-AR(1) on each subsample. AUC and correlation compare the filtered bear-regime probability with the contemporaneous dated bear indicator (in sample).

The long real bear markets of 1968 to 1982 were confirmed slowly, 11 to 16 months after their peaks (Table 2), while the declines of 2000 to 2002 and 2007 to 2009 were confirmed within 7 and 9 months. Excluding the 2007-2009 crisis (2007:10 to 2009:06) changes little (one-month skill 18.5% for persistence and 27.4% for the logit).

6.4 Additional predictors

Table 16 evaluates each of the 11 predictors as a univariate logit. The strongest single predictors at one month are price-based: the drawdown from the twelve-month high (skill 26.5%, AUC 0.852) and the twelve-month real price change (23.2%). The drawdown measures how far a decline has already gone before the 20% threshold is crossed, so it recognizes the same ongoing bear markets slightly earlier than the real-time state; at three months its advantage over persistence is no longer significant ($p = 0.131$ with the long window). The additional valuation and macroeconomic predictors add nothing at one month: the excess CAPE yield (0.1%), the payout ratio (1.3%), earnings growth (0.0%) and the level of the ten-year yield (−4.7%). At six and twelve months the

Table 14: Bear markets under alternative dating rules

Dating rule	Bear markets, 1871-2026	Since 1960	Bear share since 1960 (%)	Median decline (%)
Real prices, 15%	38	12	23.7	28.6
Real prices, 20% (main)	25	9	21.8	34.2
Real prices, 25%	20	7	21.9	38.7
Nominal prices, 20%	23	7	13.5	32.2

Notes: Symmetric thresholds: a bear market starts when the index falls the stated percentage below its peak and ends when it rises the same percentage above its trough. Nominal dating uses the S&P Composite monthly average price without deflating.

Table 15: Robustness of out-of-sample Brier skill (%) to dating, estimation window and evaluation period

	h = 1	h = 3	h = 6	h = 12
Persistence				
Main (real prices, 20%, expanding)	17.6***	10.0***	3.9**	−0.2
Dating threshold 15%	22.9***	12.0***	4.9***	−0.5
Dating threshold 25%	10.1***	5.2***	1.2	−1.0
Nominal prices, 20%	17.2***	10.3***	4.2***	−0.8
Rolling 30-year window	10.9**	2.6	−4.2	−6.3
Targets 1960-1989	8.0**	2.8	−0.4	−0.5
Targets 1990-2026	28.3***	18.0***	8.7***	0.3
Excluding 2007-2009 crisis	18.5***	10.8***	4.6**	0.0
MS-AR(1)				
Main (real prices, 20%, expanding)	9.1***	2.7	−1.4	−1.8
Dating threshold 15%	8.5***	2.0	−1.6	−1.5
Dating threshold 25%	9.7***	2.9	−2.5	−3.2
Nominal prices, 20%	13.8***	7.0***	1.4	−0.5
Rolling 30-year window	6.7**	−1.2	−4.6	−5.0
Targets 1960-1989	1.7	−4.8	−6.7	−4.6
Targets 1990-2026	17.4***	11.0***	4.4	1.1
Excluding 2007-2009 crisis	8.9***	2.4	−1.1	−1.7
Logit: dividend yield				
Main (real prices, 20%, expanding)	18.3***	13.4***	8.2**	1.4
Dating threshold 15%	13.8***	8.5***	3.2	−5.1
Dating threshold 25%	13.1***	9.6***	5.3	0.3
Nominal prices, 20%	15.4***	11.8***	6.9*	−1.4
Rolling 30-year window	3.0	−1.2	−5.2	−10.5
Targets 1960-1989	10.6***	5.1**	0.8	−3.8
Targets 1990-2026	26.7***	22.5***	16.4**	7.1
Excluding 2007-2009 crisis	21.5***	16.1***	10.4***	2.7
Logit: 6 main predictors				
Main (real prices, 20%, expanding)	25.4***	12.4***	1.7	−3.2
Dating threshold 15%	25.1***	10.2**	−0.7	−5.8
Dating threshold 25%	25.3***	13.0***	1.8	−7.2
Nominal prices, 20%	26.4***	16.4***	7.3*	4.0
Rolling 30-year window	21.2***	−2.4	−27.8	−54.7
Targets 1960-1989	17.7***	2.4	−9.4	−15.8
Targets 1990-2026	34.0***	23.4***	13.9*	10.5
Excluding 2007-2009 crisis	27.4***	14.6***	4.2	−1.2

Notes: Skill is the percentage reduction in Brier score relative to the unconditional frequency estimated in the same variant. Each dating variant re-dates bear markets (symmetric thresholds) and re-runs the complete real-time exercise, including the vintage labels and the real-time state. The rolling variant estimates every model on the most recent 30 years of training data. The evaluation-period rows score the main forecasts on subsets of target months; the crisis excluded is 2007:10-2009:06. Stars: one-sided Diebold-Mariano test against the unconditional frequency, * p < 0.10, ** p < 0.05, *** p < 0.01.

dividend yield remains the best single predictor (8.2% and 1.4%). The logit on all 11 predictors is the most accurate model at one month (29.1%) and overfits at twelve (−14.4%).

7 Limitations

The sample contains 9 bear markets since 1960, so the out-of-sample evaluation rests on few independent episodes, and the overlapping monthly forecasts are strongly dependent. The block bootstrap and the HAC-based tests account for this dependence only approximately, and the wide intervals are an honest summary of how little a sample of this size can resolve. Monthly average prices understate intra-month declines: the 2020 decline, for example, reaches only 19.6% in monthly averages and is not dated as a bear market. Publication lags are imposed conservatively, but the Shiller series are themselves revised and partly interpolated, and this paper uses a single vintage, so the exercise is quasi-real-time rather than a true vintage analysis; the comparison with

Table 16: Out-of-sample Brier skill (%) of individual predictors and combinations

	h = 1	h = 3	h = 6	h = 12	AUC, h = 1
Baseline					
Persistence	17.6***	10.0***	3.9**	−0.2	0.707
Univariate logits: main predictors					
Log dividend yield	18.3***	13.4***	8.2**	1.4	0.646
Log CAPE	13.9***	8.3***	2.7*	−3.9	0.668
CPI inflation, 12-month	5.0***	3.6***	1.9*	0.2	0.775
12-month change in 10-year yield	0.8	−0.4	−0.9	−0.5	0.664
12-month real price change	23.2***	12.2***	4.6**	0.0	0.831
12-month realized volatility	2.2**	0.5	−0.9	−1.1	0.614
Univariate logits: additional predictors					
Excess CAPE yield	0.1	−2.0	−3.7	−5.7	0.554
Log payout ratio	1.3*	0.4	−0.4	−1.2	0.504
12-month real earnings growth	0.0	−0.6	−1.0	−1.1	0.622
10-year Treasury yield	−4.7	−6.0	−7.4	−7.8	0.562
Drawdown from 12-month high	26.5***	13.8***	4.4*	−0.2	0.852
Multi-predictor forecasts					
Equal-weight combination (11 univariate logits)	13.0***	7.2***	2.7***	−0.7	0.783
DMA of the 11 univariate logits	23.3***	10.6***	2.3	−4.5	0.772
Logit: all 11 predictors	29.1***	12.8**	−3.0	−14.4	0.787

Notes: Real-time expanding-window forecasts with the main dating rule; every model is re-estimated at each origin on data available then. Skill is relative to the unconditional frequency. Stars: one-sided Diebold-Mariano test against the unconditional frequency, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the archived extract suggests that revisions are small and confined to the most recent months. The machine-learning models are re-estimated once a year with small hyperparameter grids to keep the computation modest, and richer specifications might do differently. The predictor set is limited to what Shiller's workbook contains; credit spreads and the term structure, which Chen (2009) and Chen, Chen and Chou (2017) find useful, are not in it.

8 Conclusion

The regime-switching description of the S&P 500 is robust: real returns alternate between a long, calm bull regime and a short, volatile bear regime in every specification and subsample examined here, and the switching probabilities move with inflation and recent drawdowns. The claim that market and macroeconomic indicators forecast bear markets is much weaker once forecasts are made in real time. The information that best predicts a bear market next month is that one has already begun. Logit models, machine-learning classifiers and forecast combinations built on valuation, inflation, momentum and volatility improve on that persistence benchmark by several points of Brier skill at one month, mostly through better calibration, but the improvement is not distinguishable from chance once the dependence of forecast errors and the number of models are taken into account. Beyond a few months, the dividend yield retains a small and imprecise advantage over the unconditional frequency, and the flexible models do worse than the base rate. These conclusions survive changes to the dating rule, the regime model, the estimation window, the evaluation period and the predictor set. They are consistent with a market in which downturns are difficult to anticipate from public information, and they suggest that the useful contribution of these models is faster recognition of a bear market, not advance warning of one.

References

Ang, A. and Timmermann, A. (2012). Regime changes and financial markets. *Annual Review of Financial Economics*, 4(1), 313-337.

Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1-3.

Campbell, J. Y. and Shiller, R. J. (1988). The dividend-price ratio and expectations of future dividends and discount factors. *Review of Financial Studies*, 1(3), 195-228.

Chen, N.-K., Chen, S.-S. and Chou, Y.-H. (2017). Further evidence on bear market predictability: The role of the external finance premium. *International Review of Economics & Finance*, 50, 106-121.

Chen, S.-S. (2009). Predicting the bear stock market: Macroeconomic variables as leading indicators. *Journal of Banking & Finance*, 33(2), 211-223.

Clark, T. E. and West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291-311.

Diebold, F. X., Lee, J.-H. and Weinbach, G. C. (1994). Regime switching with time-varying transition probabilities. In C. P. Hargreaves (Ed.), *Nonstationary Time Series Analysis and Cointegration* (pp. 283-302). Oxford University Press.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253-263.

Estrella, A. and Mishkin, F. S. (1998). Predicting U.S. recessions: Financial variables as leading indicators. *Review of Economics and Statistics*, 80(1), 45-61.

Fama, E. F. and French, K. R. (1988). Dividend yields and expected stock returns. *Journal of Financial Economics*, 22(1), 3-25.

Ferrari, S. and Cribari-Neto, F. (2004). Beta regression for modelling rates and proportions. *Journal of Applied Statistics*, 31(7), 799-815.

Filardo, A. J. (1994). Business-cycle phases and their transitional dynamics. *Journal of Business & Economic Statistics*, 12(3), 299-308.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189-1232.

Giacomini, R. and White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545-1578.

Goetzmann, W. N. and Jorion, P. (1993). Testing the predictive power of dividend yields. *Journal of Finance*, 48(2), 663-679.

Gu, S., Kelly, B. and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223-2273.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357-384.

Hammerschmid, R. and Lohre, H. (2018). Regime shifts and stock return predictability. *International Review of Economics & Finance*, 56, 138-160.

Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4), 365-380.

Hansen, P. R., Lunde, A. and Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2), 453-497.

Harvey, D., Leybourne, S. and Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281-291.

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3), 1217-1241.

Lewellen, J. (1999). The time-series relations among expected return, risk, and book-to-market. *Journal of Financial Economics*, 54(1), 5-43.

Lunde, A. and Timmermann, A. (2004). Duration dependence in stock prices: An analysis of bull and bear markets. *Journal of Business & Economic Statistics*, 22(3), 253-273.

Maheu, J. M. and McCurdy, T. H. (2000). Identifying bull and bear markets in stock returns. *Journal of Business & Economic Statistics*, 18(1), 100-112.

Murphy, A. H. (1973). A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4), 595-600.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703-708.

Pagan, A. R. and Sossounov, K. A. (2003). A simple framework for analysing bull and bear markets. *Journal of Applied Econometrics*, 18(1), 23-46.

Pontiff, J. and Schall, L. D. (1998). Book-to-market ratios as predictors of market returns. *Journal of Financial Economics*, 49(2), 141-160.

Raftery, A. E., Kárný, M. and Ettler, P. (2010). Online prediction under model uncertainty via dynamic model averaging: Application to a cold rolling mill. *Technometrics*, 52(1), 52-66.

Rapach, D. E., Strauss, J. K. and Zhou, G. (2010). Out-of-sample equity premium prediction: Combination forecasts and links to the real economy. *Review of Financial Studies*, 23(2), 821-862.

Shiller, R. J. (2015). *Irrational Exuberance* (3rd ed.). Princeton University Press. Data: U.S. Stock Markets 1871-Present and CAPE Ratio, shillerdata.com, downloaded 18 October 2026.

Smithson, M. and Verkuilen, J. (2006). A better lemon squeezer? Maximum-likelihood regression with beta-distributed dependent variables. *Psychological Methods*, 11(1), 54-71.

Stambaugh, R. F. (1999). Predictive regressions. *Journal of Financial Economics*, 54(3), 375-421.

Welch, I. and Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455-1508.

White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097-1126.

Zou, H. and Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B*, 67(2), 301-320.

A Regressions of the regime probability and the award version

The award version of this thesis forecast the filtered regime probability Q_t rather than a dated event, with univariate OLS and beta regressions (Ferrari and Cribari-Neto, 2004; Smithson and Verkuilen, 2006) of Q_{t+h} on each of 7 monthly-change predictors,

$$Q_{t+h} = \alpha + \beta x_t + e_{t+h}, \tag{5}$$

and an out-of-sample comparison with an intercept-only model. Table 17 re-estimates this design on the Shiller data with predictors observable at t, growth rates in place of index levels and Newey-West standard errors. The number of predictors significant at the 5% level is 4 of 7 at one month, 3 at three, 6 at six and 1 at twelve. The largest R^2 (0.125) belongs to the real total return at one month (coefficient -9.37^{***} (1.06) percentage points), which is close to mechanical because the filter itself updates on recent returns. Total-return-scaled earnings show no relationship (R^2 = 0.001). Because Q_t uses full-sample parameters, these are in-sample associations rather than forecasts.

Table 19 reports the out-of-sample comparison. Panel C uses expanding-window refits from April 2011: 17 of 28 OLS ratios of mean squared prediction errors fall below one and 14 are significant at the 10% level by the Clark and West (2007) test (12 with 23 Bartlett lags), with a best ratio of 0.850. The regime probability's own current value achieves 0.468 at one month and no gain at six months (1.059).

The award version reached stronger conclusions for three reasons that a replication of its code path on its own data confirms (Table 18, Figure 10): its regime model was estimated with the following month's price as a regressor, its "k months ahead" targets were shifted backward rather than forward in time, and its loss ratio compared squared mean errors rather than mean squared

Table 17: Corrected award-version design: regime probability h months ahead on each predictor

	OLS (percentage points)				Beta regression (logit link)			
	h = 1	h = 3	h = 6	h = 12	h = 1	h = 3	h = 6	h = 12
CPI inflation, monthly	3.773^{**}	4.037^{*}	4.296^{**}	4.326^{***}	0.162^{**}	0.200^{**}	0.173^{**}	0.174^{***}
	(1.741)	(2.206)	(1.694)	(1.490)	(0.073)	(0.093)	(0.069)	(0.059)
R^2 / pseudo-R^2	0.020	0.023	0.026	0.026	0.022	0.027	0.027	0.030
Change in 10-year yield	−0.046	−1.211	3.718^{***}	0.948	0.019	−0.048	0.135^{***}	0.033
	(1.291)	(1.061)	(1.016)	(0.790)	(0.059)	(0.039)	(0.043)	(0.027)
R^2 / pseudo-R^2	0.000	0.002	0.020	0.001	0.000	0.002	0.018	0.001
Real bond return	−0.117	0.932	-4.021^{***}	−1.361	−0.020	0.035	-0.148^{***}	−0.048
	(1.359)	(1.250)	(1.179)	(0.879)	(0.059)	(0.047)	(0.048)	(0.030)
R^2 / pseudo-R^2	0.000	0.001	0.023	0.003	0.000	0.001	0.021	0.003
Real dividend growth	-3.525^{***}	-3.054^{**}	-3.341^{**}	−2.661	-0.114^{**}	-0.103^{**}	-0.113^{**}	−0.107
	(1.296)	(1.318)	(1.366)	(1.972)	(0.046)	(0.044)	(0.046)	(0.071)
R^2 / pseudo-R^2	0.018	0.013	0.016	0.010	0.014	0.012	0.014	0.013
Real total return	-9.366^{***}	-6.142^{***}	-5.482^{***}	−1.250	-0.417^{***}	-0.241^{***}	-0.208^{***}	−0.039
	(1.056)	(1.071)	(1.268)	(0.950)	(0.044)	(0.046)	(0.051)	(0.032)
R^2 / pseudo-R^2	0.125	0.054	0.043	0.002	0.110	0.049	0.039	0.002
TR-scaled earnings growth	−0.678	−1.527	−2.073	-2.493^{*}	−0.070	−0.061	−0.066	-0.080^{*}
	(2.441)	(1.748)	(1.625)	(1.282)	(0.124)	(0.071)	(0.062)	(0.048)
R^2 / pseudo-R^2	0.001	0.003	0.006	0.009	0.002	0.003	0.005	0.006
Change in log TR-CAPE	-9.299^{***}	-6.133^{***}	-5.560^{***}	−1.437	-0.414^{***}	-0.242^{***}	-0.210^{***}	−0.045
	(1.053)	(1.088)	(1.264)	(0.978)	(0.043)	(0.047)	(0.051)	(0.032)
R^2 / pseudo-R^2	0.124	0.054	0.044	0.003	0.109	0.049	0.040	0.002
Observations	796	794	791	785	796	794	791	785

Notes: Each cell is a separate univariate regression of the filtered bear-regime probability Q(t+h) from the MS-AR(1) model on the standardized predictor at t. OLS coefficients are in percentage points of probability. Newey-West standard errors in parentheses (7, 12 lags by horizon). Forecast origins 1960:02 to 2026:05 for h = 1. The beta-regression pseudo-R^2 is the squared correlation of the linear predictor with logit(Q). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-sided). Q uses full-sample parameters, so these are in-sample associations, not forecasts.

Table 18: Award-version results, their replication, and the corrected estimates

	Award version (reported)	Replicated	Corrected
P(bull to bull)	0.945	0.950	0.940
P(bear to bear)	0.833	0.833	0.798
Dividend, k = 1: coefficient	−0.195	−0.198	−3.52*** (1.30) pp
Dividend, k = 6: R^2	0.075	0.074	0.016
Total return, k = 1: R^2	0.108	0.107	0.125
Dividend, k = 1: OLS forecast ratio	0.004	0.029	0.973
Total return, k = 1: OLS forecast ratio	0.014	0.006	0.852

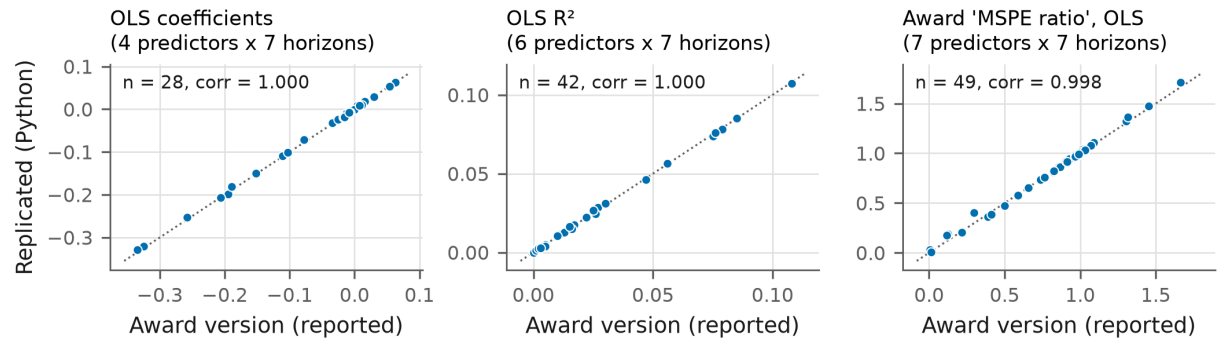
Notes: Reported values are transcribed from the award PDF (v2/data/archived_reported.csv). 'Replicated' re-runs the award version's code path on one45.csv, including the future-price regressor, the backward target Q(t-k) and its loss ratio. 'Corrected' uses the MS-AR(1) model on returns from the Shiller workbook, past-only predictors, the forward target Q(t+k) and expanding-window forecasts. Corrected coefficients are per one-standard-deviation change of a growth-rate transformation and are comparable in sign and significance only.

errors. The replication reproduces the reported transition probabilities (0.950 and 0.833 against 0.945 and 0.833) and the reported coefficients (correlation 1.000). For dividends at one month the reported ratio of 0.004 becomes 1.154 as a true MSPE ratio on the same design and 0.973 after all corrections. Panels A and B of Table 19 show the award version's design with its own ratio and with the correct one.

Table 19: Forecasting the regime probability: unrestricted versus intercept-only models

	OLS				Beta regression			
	h = 1	h = 3	h = 6	h = 12	h = 1	h = 3	h = 6	h = 12
A. Award-version design and formula (squared mean error ratio)								
CPI	0.923	0.735	0.868	0.925	0.947	0.904	0.961	0.993
Long-term rate	0.981	0.975	1.008	0.997	0.992	0.981	1.002	0.996
Real bond	0.818	0.862	1.077	0.948	0.960	0.949	1.014	0.959
Real dividend	0.029	0.189	0.404	0.179	0.370	0.275	0.210	0.276
Real total return	0.006	0.921	0.942	0.982	0.564	1.019	0.956	1.021
TR-scaled earnings	0.363	0.474	0.759	1.365	0.819	0.837	0.894	1.222
TR-CAPE	0.822	1.016	1.018	1.027	0.706	1.010	1.004	1.013
B. Award-version design, correct mean squared prediction error ratio								
CPI	1.002	1.021	1.005	1.006	0.996	0.996	0.997	1.000
Long-term rate	0.989	1.009	1.002	1.001	0.991	1.002	1.001	1.002
Real bond	0.997	1.041	1.019	1.011	0.970	1.027	1.006	1.015
Real dividend	1.154	1.184	1.148	0.963	0.951	0.924	0.883	0.806
Real total return	1.539	1.010	1.003	1.002	0.770	1.001	1.000	0.999
TR-scaled earnings	0.862	0.879	0.946	0.965	0.799	0.861	0.945	1.003
TR-CAPE	0.720	0.994	0.998	0.990	0.664	0.996	1.000	0.994
C. Corrected design: past-only predictors, true h-month lead, expanding window								
CPI inflation, monthly	0.993	0.958**	0.946**	0.942***	0.972**	0.936***	0.939***	0.938***
Change in 10-year yield	1.003	1.001	0.979*	1.005	1.004	0.999	0.987	1.003
Real bond return	1.007	1.003	0.981	1.010	1.009	0.998	0.994	1.009
Real dividend growth	0.973**	0.983*	0.950***	0.958**	0.951***	0.959***	0.943***	0.948***
Real total return	0.852***	0.925***	0.998*	1.003	0.826***	0.931***	0.981**	0.999
TR-scaled earnings growth	0.999	1.005	1.022	1.049	0.997	1.002	1.010	1.023
Change in log TR-CAPE	0.850***	0.924***	0.997*	1.006	0.833***	0.934***	0.985**	1.001
Benchmark: regime probability at t	0.468***	0.914***	1.059	1.021	0.479***	0.888***	1.006	1.004

Notes: Entries are ratios of the unrestricted model's loss to the intercept-only model's loss; values below one favor the predictor. Panels A and B re-run the award version's code path on one45.csv (target Q(t-k), regime model with the next month's price, single 80/20 split with the test period starting in 2011). Panel A uses its formula (mean error)² / (mean error)², which is not a mean squared error. Panel C forecasts Q(t+h) from predictors observable at t with expanding-window refits on the Shiller data, origins 2011:04 to 2026:05 for h = 1. The benchmark row uses Q(t) itself as the predictor. Stars in Panel C mark Clark-West (2007) tests: * p < 0.10, ** p < 0.05, *** p < 0.01.



Source: R. J. Shiller, U.S. Stock Markets 1871-Present and CAPE Ratio (shillerdata.com, downloaded 18 October 2026); author's calculations.

Figure 10: Values reported in the award version against a Python replication of its code path on its archived data, including its timing errors.

B Data source and vintage

The workbook used here is `ie_data.xls`, downloaded on 18 October 2026 from the download link on shillerdata.com (1.7 MB, server last-modified date 6 October 2026). Its SHA-256 hash is shown below, and `thesis.py` refuses to run on a file with a different hash, so a future Shiller update has to be adopted deliberately.

`98a54927942e842ec07ffe73000998e793953708c345b6b27bb363e9ada8366a`

The parser reads the "Data" sheet, checks each column heading, converts Shiller's fractional dates (1871.01 is January 1871 and 1871.1 is October), drops the note rows below the table and leaves blank recent cells missing. Table 20 compares the archived extract `one45.csv` with this vintage month by month; a list of every differing observation is in `output/results/one45_differences.csv`.

Table 20: The archived extract one45.csv against the 6 October 2026 Shiller workbook

Series	Overlap	Months	Months that differ	Largest difference (%)	Months affected
S&P Composite price	1900:01-2024:03	1,491	1	0.65	2024:03-2024:03
Dividends	1900:01-2023:12	1,488	0		
Earnings	1900:01-2023:12	1,488	3	0.22	2023:10-2023:12
CPI	1900:01-2024:03	1,491	2	0.85	2024:02-2024:03
Long-term interest rate	1900:01-2024:03	1,491	1	0.95	2024:03-2024:03
CAPE	1900:01-2024:03	1,491	2	0.67	2024:02-2024:03
Total-return CAPE	1900:01-2024:03	1,491	3	0.47	2024:01-2024:03
Real price (rescaled)	1900:01-2024:03	1,491	2	0.35	2024:02-2024:03
Real TR-scaled earnings (rescaled)	1900:01-2023:12	1,488	3	0.22	2023:10-2023:12

Notes: A month differs if the archived value differs from the current value rounded to two decimals by more than 0.01 (levels), or by more than 0.05% after rescaling (real series, which Shiller expresses in dollars of the latest month). 'Largest difference' is the largest absolute percentage difference among the months that differ. All differences fall in the last months of the archived file, which held estimates at the time (see output/results/one45_differences.csv).